# Software Requirements Specification(SRS)

# Project:- Vishvin’s Enterprise Resource Planning

# A/c: Vishvin’s Management Team

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# Introduction

## 1.1 Overview of the Requirement Specifications Document

This Requirement Specifications Document describes the **functional and non-functional requirements** of the **Vishvin's Enterprise Resource Planning (VERP).** The document serves as a formal reference for ~~stakeholders,~~ project managers, designers, developers, testers, and implementation teams involved in the development and deployment of the system.

### 1.1.1 Purpose of the Document

- Clearly define the **scope and objectives** of the VERP
- Specify **business rules, workflows, and system behaviour**
- Act as a baseline for **design, development, testing, and acceptance**
- Ensure a common understanding between **business users and technical teams**

## The document Defines:

**a. Defines Requirements:**

- **Functional:** Covers end-to-end **Enterprise Resource Planning (ERP) workflows**, including user and role management, master data management, procurement, inventory management, sales, purchase requisitions, approvals, financial transactions, reporting, and other business operations as applicable to Vishvin ERP. The system includes centralized data management, role-based access, workflow automation, dashboards, and reports to support efficient business operations and decision-making.
- **Non-Functional:** Specifies system **performance** (efficient processing of business transactions and reports), **security** (role-based access control, secure authentication, and encrypted data transmission), **scalability** (supporting business growth and increasing transaction volumes), **usability** (simple and user-friendly interfaces), **reliability** (consistent system availability and data integrity), and **maintainability** (ease of system updates, enhancements, and support).

**b. Guides Implementation:**

- Acts as a **contractual and technical blueprint** for system integrators and vendors to design, develop, and deploy the **Workforce Management System (WFMS)**, including **web applications, mobile applications, backend services, and database components**.
- Provides clear guidance to ensure the solution adheres to **defined business workflows, role-based access controls, data security practices, audit logging, and performance standards**, as approved by the organization and governing authorities.
- Ensures compliance with applicable **government IT standards, data protection policies, hosting guidelines, and audit requirements**, including secure infrastructure, user activity tracking, and system logging as mandated.

**c. Ensures Accountability:**

- Facilitates clear accountability and transparency between **Vishvin management, internal stakeholders, and implementation teams** by documenting business workflows, operational rules, user responsibilities, approval processes, and acceptance criteria for the Vishvin ERP system.

- Establishes a **single source of truth** for system requirements, supporting future audits, operational reviews, system maintenance, and enhancements through **version-controlled documentation, defined change management processes, and requirement traceability**.

### 1.1.2 Distribution List

The following stakeholders and team members are intended recipients of this Software Requirements Specification (SRS) document. Each individual or group listed plays a critical role in the development, review, approval and implementation of the Workforce Management System for ANG.

| Sl.No | Department / Organization | Name / Designation | Role |
|---|---|---|---|
| 1 | Vishvin Technologies Pvt Ltd., | MD | Approver |
| 2 | IT / Project Manager | Vishvin Technologies Pvt. Ltd. | Project Lead / Developer Liaison |
| 3 | Software Development Team | | Implementation Team |
| 4 | Quality Assurance Team | | Testing Team |

### 1.1.3 Current System

The existing business operations at **Vishvin Technologies Pvt. Ltd.** are predominantly manual, relying on hardcopy documents, physical registers, and manual record-keeping for managing day-to-day activities. Business information is maintained across separate records, making data retrieval, tracking, reporting, and coordination time-consuming.

### 1.1.4 Limitations of the Current System

The current manual system has the following limitations:

- **Manual Documentation:** Heavy dependency on hardcopy records and physical registers.
- **Lack of Centralization:** Business information is scattered across different records and departments.
- **Limited Visibility:** Difficulty in tracking operational activities and accessing updated information.
- **Time-Consuming Processes:** Manual data entry, document retrieval, and report preparation.
- **Data Accuracy:** Increased risk of errors, duplication, and loss of records.
- **Limited Scalability:** Managing increasing business operations becomes difficult with manual processes.

## 1.2 Scope of the Project

**Vishvin ERP (Enterprise Resource Planning)** is a centralized web-based software platform designed to streamline and manage organizational activities through integrated administration, procurement, and daily reporting modules.

The system aims to digitize existing manual and hardcopy-based processes, centralize business information, standardize workflows, and improve operational visibility, accountability, and reporting across the organization.

### A. Project Scope Includes

- The project shall cover the design, development, testing, deployment, and maintenance of the Vishvin ERP software within the approved scope.
- The **Super Admin Module** shall support centralized administration, company and product management, project masters, designations, roles and permissions, user management, and material-related master data through the currently defined 12 screens.
- The **Purchase Requisition and Purchase Order (PR–PO) Module** shall support procurement-related activities, including purchase requisition creation, approval workflows, quotation tracking, vendor management, purchase order generation, document tracking, and procurement reporting.
- The **Daily Reporting Module** shall support employee daily work entry, draft saving, submission, manager review, team activity monitoring, department-wise reporting, management reports, and audit trail functionality.
- The system shall provide appropriate role-based access to users according to their assigned responsibilities and permissions.
- The system shall centralize relevant business records and provide dashboards, reports, and status tracking for supported activities.
- The system shall support future expansion through additional modules and functionalities based on approved business requirements and change requests.

### B. Performance Requirements:

- The system shall provide responsive access to supported screens, dashboards, and reports under normal operating conditions.
- Data retrieval, transaction processing, and report generation shall perform efficiently for the expected workload.
- Specific response-time targets and concurrent-user capacity shall be defined based on agreed business and technical requirements.

### C. Security Requirements:

The system shall implement appropriate security controls to protect organizational and business information.

- Role-based access control according to assigned user roles and permissions.
- Secure user authentication and access management.
- Validation of user inputs and authorized business operations.
- Secure data transmission using HTTPS/TLS.
- Appropriate protection of stored data and prevention of unauthorized access.

### D. Architectural & Quality Attributes

- **Modularity:** The system shall support organized functional modules and future module additions.
- **Maintainability:** The software shall be structured to facilitate troubleshooting, updates, and enhancements.
- **Scalability:** The system shall accommodate increasing users, data volumes, and business requirements.
- **Reliability:** The system shall maintain data integrity and provide appropriate backup and recovery mechanisms.
- **Usability:** The interfaces shall be clear, consistent, and easy for authorized users to operate.
- **Extensibility:** Additional ERP modules and integrations may be introduced through approved future phases.

## E. Key Functionalities Include:

The current scope of Vishvin ERP includes the following functional areas:

**1. Super Admin Module**

- Company, product, project, and material master management.
- Designation and user management.
- Role and permission management.

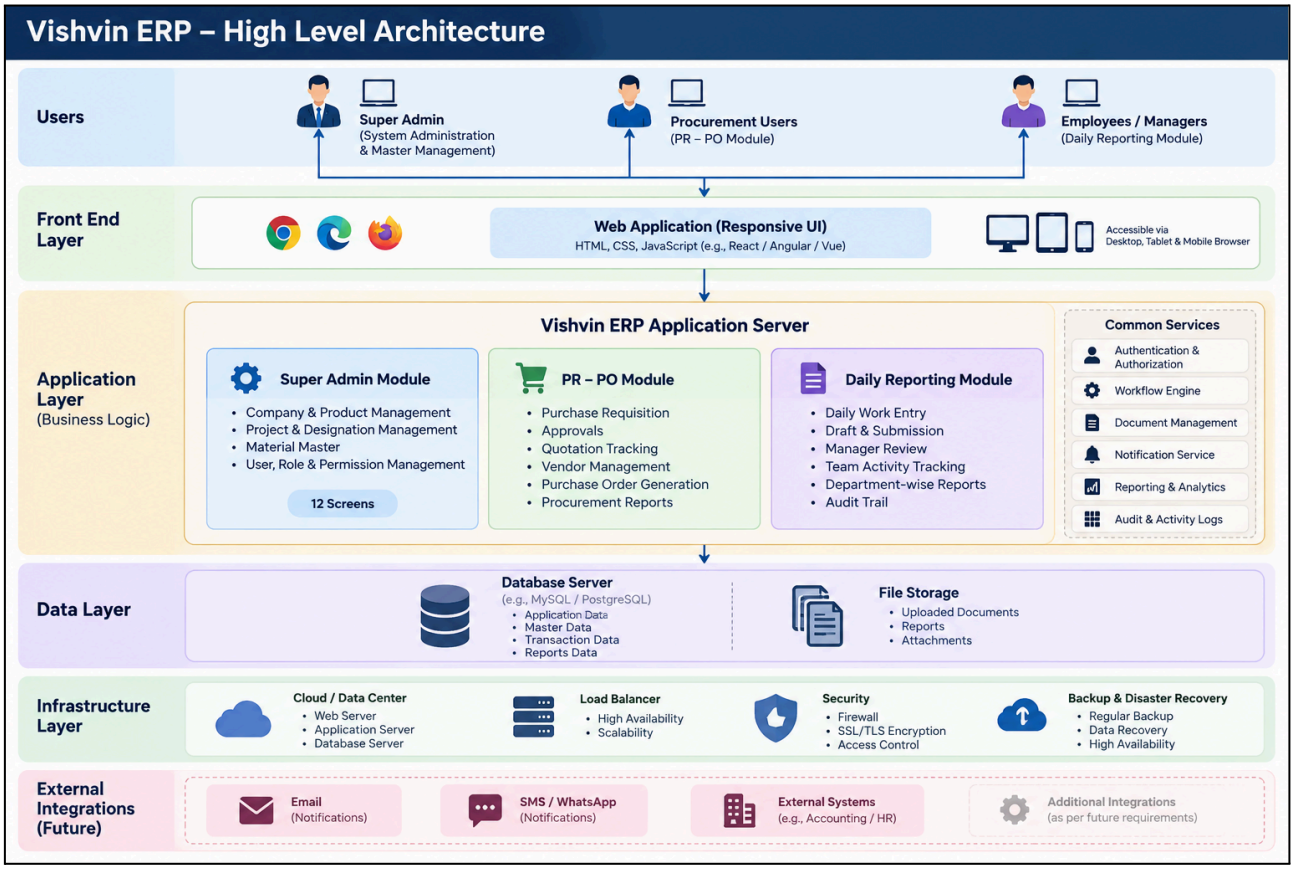
**2. Purchase Requisition and Purchase Order (PR–PO) Module**

- Purchase requisition creation and approval.
- Quotation tracking and vendor management.
- Purchase order generation and tracking.
- Document management and procurement reporting.

**3. Daily Reporting Module**

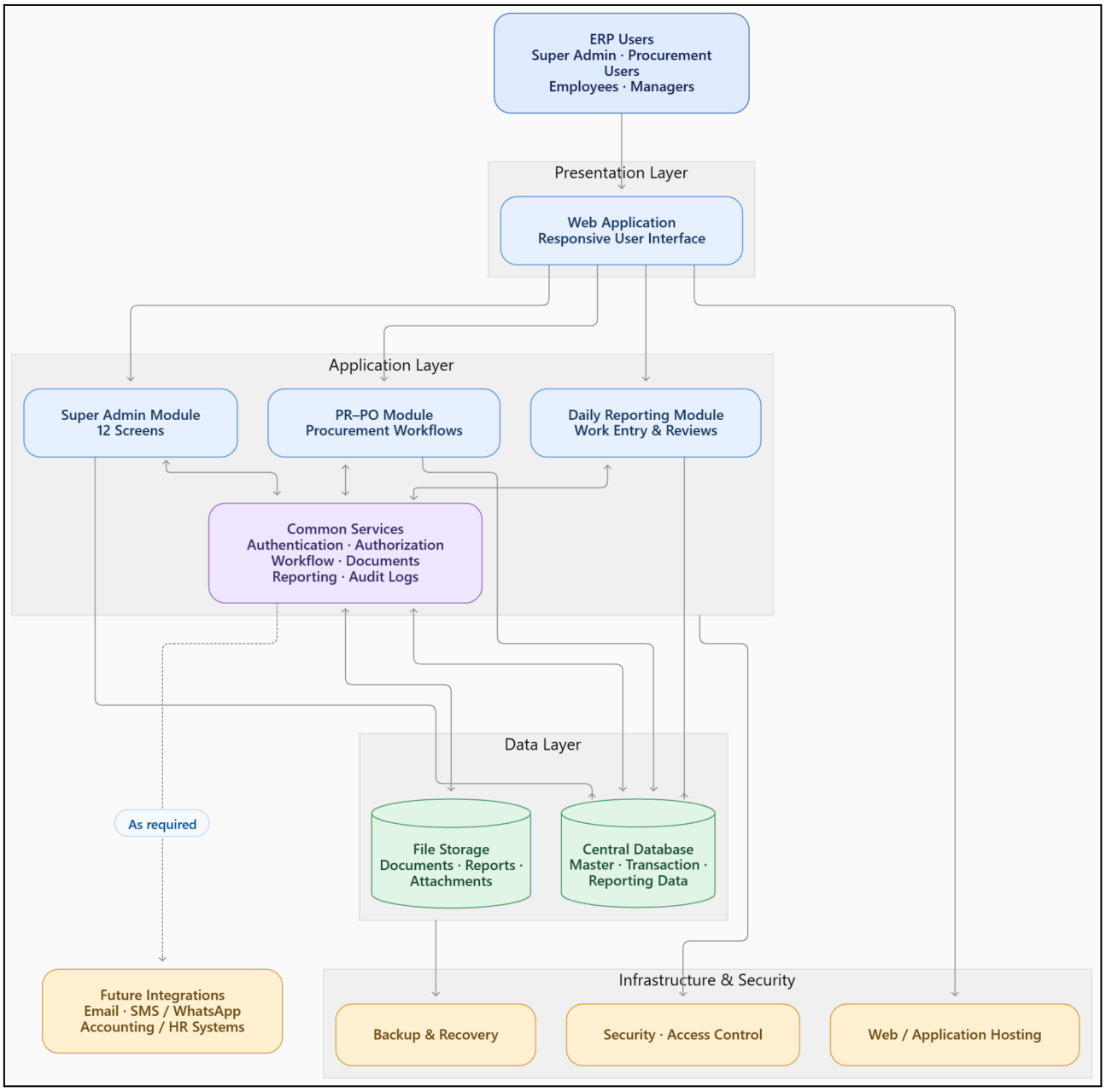
- Employee daily work entry and submission.
- Draft management and manager review.
- Team activity monitoring and departmental reporting.
- Management reports and audit trail.

### 1.2.1 Workforce Management System- High Level Architecture:



## A. Architectural Goals

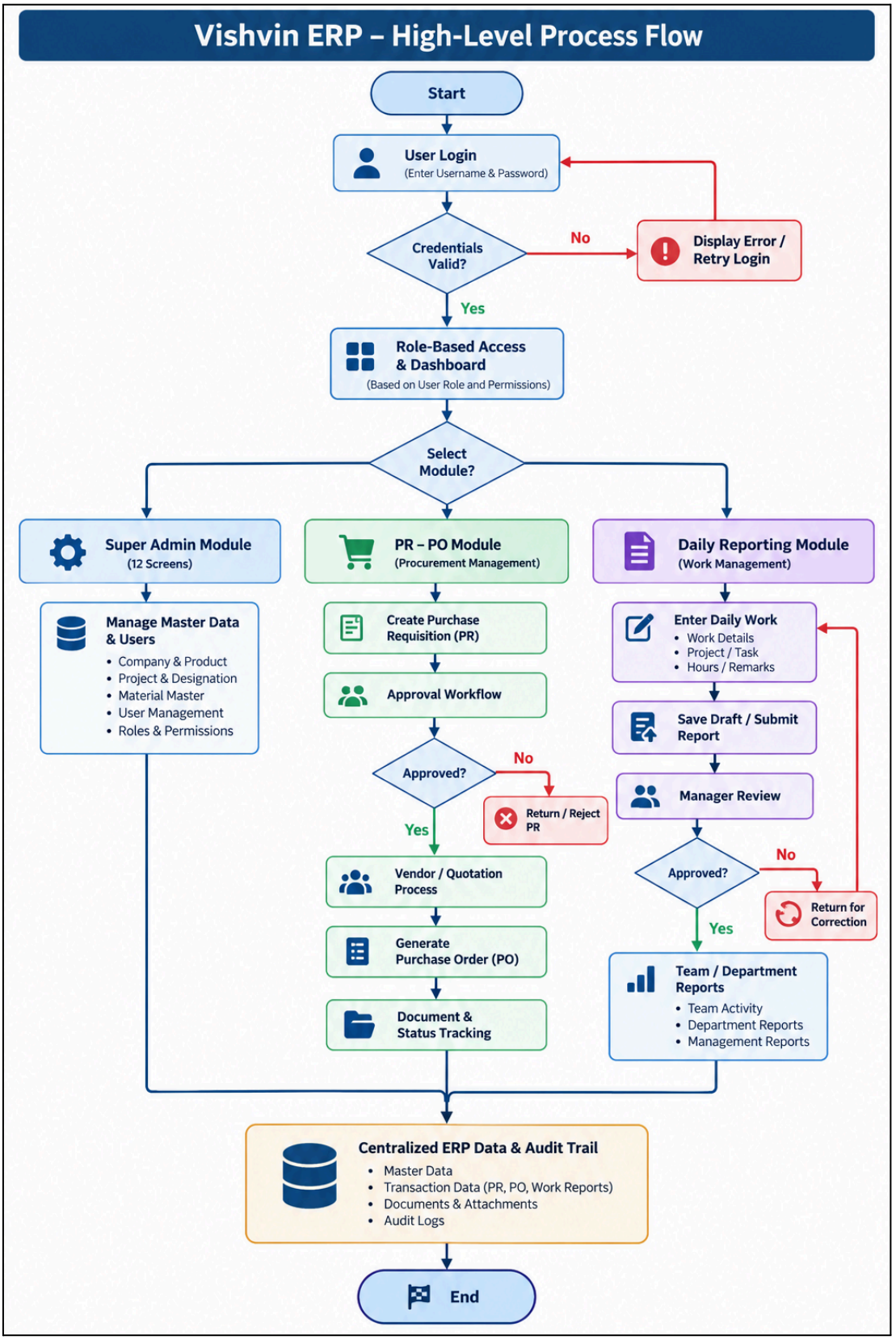
The Vishvin's Enterprise Resource Planning (VERP) shall be designed to achieve scalability, performance, security, reliability, maintainability, interoperability, and flexibility. Detailed measurable requirements for these quality attributes are specified in Section 3.6 – Non-Functional Requirements, while implementation and architectural decisions are documented separately in the Design Document (DD).




## B. WFMS High-Level Architecture Components:

Vishvin ERP follows a modular, role-based architecture comprising a web-based user interface, centralized application services, database management, secure infrastructure, and support for future external integrations. The architecture currently covers the **Super Admin, PR–PO, and Daily Reporting modules**, supported by common services for authentication, authorization, workflow management, document handling, reporting, and audit tracking. The high-level architecture is illustrated below. Detailed architectural design shall be documented separately in the Solution Architecture Document (SAD), where applicable.

## C. VERP Process Flow



# 2 Application Environment

## 2.1 Business Context

Vishvin ERP is designed to support and streamline the organization's business operations through a centralized web-based platform. The system currently covers three key functional areas: **Super Admin, Purchase Requisition and Purchase Order (PR–PO), and Daily Reporting**.

The application aims to reduce dependency on manual documentation, improve coordination between departments, standardize business workflows, and provide centralized access to business information, operational status, and management reports. The system is designed to support future expansion through additional ERP modules as business requirements evolve.

### Business Objectives / Strategic Alignment

Vishvin ERP supports the organization's digital transformation by centralizing business operations, standardizing workflows, and improving operational transparency and reporting across departments.

The system aligns with enterprise digitization initiatives by:

- Reducing dependency on manual documentation and record-keeping.
- Improving operational accountability and workflow management.
- Centralizing business data for easier access and tracking.
- Improving visibility through dashboards and management reports.
- Supporting future business expansion through additional ERP modules.

### Business Functions to be Addressed:

**1. System Administration and Master Data Management**

- Centralized management of company, product, project, designation, and material-related master data.
- User creation and management with role-based access permissions.
- Standardized maintenance of organizational and business information.

**2. Purchase Requisition and Purchase Order Management**

- Creation and tracking of purchase requisitions.
- Management of purchase requisition approval workflows.
- Vendor management and quotation tracking.
- Purchase order generation and status tracking.
- Procurement document management and reporting.

**3. Daily Reporting and Work Management**

- Recording and maintaining employee daily work entries.
- Saving work entries as drafts and submitting reports for review.
- Managerial review and tracking of submission status.
- Monitoring team activities and department-wise performance.
- Generating management reports and maintaining audit trails.

**4. Reporting and Operational Visibility**

- Providing dashboards and reports for supported business activities.
- Enabling users and management to access relevant operational information.

- Supporting business monitoring and decision-making through centralized data.

**5. Future Expansion**

- Supporting the addition of new business modules and functionalities based on evolving organizational requirements and approved changes.

### 2.1.4 User Profiles

Vishvin ERP serves multiple user categories across the organization involved in system administration, procurement, and daily work reporting. Each user shall have defined roles, access permissions, and functional responsibilities. The application shall enforce Role-Based Access Control (RBAC) to ensure users can access only the features and data authorized for their assigned roles.

| User / Role | Functional Area | Needs / Expectations |
|---|---|---|
| **Super Admin** | System Administration and Master Management | Create and manage users; maintain company, product, project, designation, and material masters; configure roles and permissions; manage access to system modules and screens. |
| **Procurement User** | Purchase Requisition and Purchase Order Management | Create purchase requisitions; track requisition status; manage quotations and vendor-related activities; view relevant purchase orders and procurement documents according to assigned permissions. |
| **Approver / Authorized Manager** | Approval and Review Workflows | Review submitted purchase requisitions; approve, reject, or return requests for correction according to assigned authority; monitor pending approvals and related statuses. |
| **Employee** | Daily Reporting | Record daily work activities; save entries as drafts; submit daily work reports; view personal submission status and feedback. |
| **Reporting Manager** | Daily/Weekly Work Review and Reporting | Review submitted employee work reports; return entries for correction where permitted; monitor team activities; access department-wise and management reports according to assigned permissions. |
| **Management User** | Management Reporting and Monitoring | View relevant dashboards, procurement summaries, daily reporting metrics, and management reports based on assigned access permissions. |

## 2.2 Volumetric Data

The Volumetric Data section outlines the expected data volumes, transaction loads, and usage patterns for **Vishvin ERP (VERP)**. The system is initially intended to support the organization's current workforce of approximately **500 employees**, with the flexibility to accommodate future growth in users, business transactions, and data volumes as additional ERP modules are introduced.

### 2.2.1 Data Volumes

**Current & Planned Scale**

- The initial deployment is expected to support approximately **500 employees** across the currently defined ERP modules.
- The system shall maintain master data, user records, purchase requisitions, purchase orders, daily reports, supporting documents, and related transaction records.
- Data volumes are expected to grow as business transactions increase and additional ERP modules are introduced.
- Database and document storage requirements shall be determined based on actual usage, transaction volumes, and retention requirements.

**Estimated Data Growth at Full Scale (Up to 2 Cr Candidates):** The initial storage capacity shall be planned according to the organization's current requirements and expected business growth. Storage allocation may be increased as the volume of ERP transactions, documents, reports, and historical records grows.

### 2.2.2 Retention Period

- Yet to Confirm

### 2.2.3 Archival Strategy

- **Yet to Confirm**

### 2.2.4 Transaction Rates

- Yet to Confirm

### 2.2.5 Scalability & Capacity Planning

- Yet to Confirm

## 2.3 Constraints

Constraints define the limitations and restrictions that influence the design, development, deployment, and operation of **Vishvin ERP (VERP)**. These constraints may arise from the operating environment, technology choices, security requirements, and organizational policies, establishing the boundaries within which the system must function.

### 2.3.1 Environment Imposed

This section describes the non-functional constraints imposed by the operating environment, infrastructure policies, and statutory requirements under which the WFMS must operate.

**a. Operating Environment**

- Vishvin ERP shall be deployed and operated on on-premises infrastructure at Vishvin Technologies Pvt. Ltd.
- The application shall be accessible through supported web browsers on authorized user devices within the configured network environment.
- The application shall support the latest stable versions of the following web browsers:
    - Google Chrome
    - Microsoft Edge
    - Mozilla Firefox
- Server operating system, hardware specifications, network configuration, and other infrastructure requirements shall be based on the actual deployment setup.

**b. Development Platform Constraints**

- Vishvin ERP shall be developed using appropriate technologies based on the project's functional requirements, maintainability, and licensing considerations.
- The application shall support integration with external systems through APIs, where required and approved as part of the project scope.
- The technology stack shall be finalized based on the actual implementation and agreed project requirements.
- Any future technology upgrades or integrations shall be evaluated based on compatibility with the existing system architecture and deployment environment.

**c. Implementation Constraints**

- Vishvin ERP shall be implemented as a browser-based web application for authorized users.
- The application shall operate within the configured on-premises hosting and network environment.
- Any future enhancements or modules shall be compatible with the existing system architecture and deployment environment.

**d. IT Policy and Security Constraints**

- Data Transmission Security: Communication between users and the application shall be secured using HTTPS/TLS, where applicable to the deployment.
- Access Control: Role-Based Access Control (RBAC) shall restrict access to modules and functionalities according to assigned user permissions.
- Data Protection: Appropriate measures shall be implemented to protect business information against unauthorized access, modification, or disclosure.
- Audit and Logging: Relevant system activities and data modifications shall be logged where supported by the application.
- Third-Party Integrations: External tools, APIs, or services shall be integrated only when required and approved as part of the project scope.

**e. Regulatory Constraints**

- Vishvin ERP shall comply with applicable legal, regulatory, and organizational requirements relevant to its operations and data handling.
- Data protection, access control, and record retention shall follow the organization's approved policies and applicable legal requirements.
- Any security audits or assessments shall be conducted as required by the organization or applicable regulations.

**f. Technical Limitations**

- System performance, scalability, and availability shall depend on the capacity of the on-premises infrastructure and the reliability of the organization's network.

- Application performance may be affected by server resources, network connectivity, and increasing data volumes.
- Future enhancements, integrations, and technology upgrades shall be compatible with the existing system architecture, security requirements, and organizational policies.

### 2.3.2 Statutory Requirements

This section outlines the legal and statutory constraints applicable to the development, deployment, and operation of the **Workforce Management System (WFMS)**. The system shall comply with all applicable **Central and State Government regulations in India**, based on the jurisdictions in which the WFMS is deployed and operated.

**a. Data Privacy and Protection –**

- The system shall handle business and employee information in accordance with applicable data protection and privacy requirements.
- Access to system data shall be controlled through assigned user roles and permissions.
- Appropriate safeguards shall be implemented to prevent unauthorized access, modification, disclosure, or loss of information.
- Data retention and deletion policies shall be defined according to organizational requirements and applicable legal obligations.

**b. Digital Evidence Handling**

- The VERP shall ensure **tamper-proof storage** of all digital evidence generated or captured through the system
- Integrity of digital evidence shall be maintained using **secure hashing mechanisms and/or digital signatures** to prevent unauthorized modification.
- The system shall implement **automatic timestamping** for:
  - Data creation
  - Upload events
  - Modifications and updates
    ensuring accurate and verifiable timelines.

**c. Audit and Record Retention**

- Record Retention Period:
  - Yet to confirm
- Historical Data Preservation:
  - Yet to confirm
- Audit Trails:
  - The VERP shall maintain immutable, time-stamped audit logs for all critical system actions, including:
    - PO generation and Approval
    - Material & Vendor Creation
    - Quotations
    - Access, modification, or export of data
  - Audit logs shall be tamper-proof and non-editable.
- Traceability for Governance:
  - Yet to confirm
- Secure Storage:
  - Yet to confirm
- Search, Export, and Reporting:
  - Yet to confirm

**d. Public Sector Software Guidelines**

- Yet to confirminfrastructure guidelines

### 2.3.3 Performance

- Yet to confirm

**b. Network Capacity Constraints**

- Yet to confirm

**a. Integration Performance Limits**

- Yet to confirm

**b. Interface to Real-Time Systems**

- Yet to confirm

**e. Target Response Times**

- Yet to confirm

# 3 Requirements Specifications

## 3.1 Details of requirements gathering process

The requirements for **VERP** were gathered through a structured and collaborative process involving key business stakeholders, operational teams, and technical representatives. The objective of the requirement's gathering process was to ensure that the system addresses real operational challenges, aligns with organizational objectives, and supports scalable workforce operations across multiple regions.

**The requirements gathering process included the following activities:**

**a. Management Guidance and Input:** The management team provides direction on business needs, operational priorities, and expected system functionality. These inputs serve as the basis for defining and refining ERP requirements.

**b. Current State Analysis:** Existing manual and hardcopy-based business processes are reviewed to identify operational challenges, information management needs, and opportunities for digitization.

**c. Requirement Elicitation:** Functional requirements are identified based on management inputs, existing HTML screens, and the intended functionality of the Super Admin, PR–PO, and Daily Reporting modules.

**d. Requirement Analysis and Validation:** The identified requirements are reviewed for clarity, feasibility, consistency, and alignment with the agreed project scope. Clarifications are obtained from the management team wherever necessary.

**e. Scope Definition and Prioritization:** Requirements are organized according to the currently defined ERP modules and implementation priorities. Additional modules and enhancements may be included in future phases based on management direction and approved requirements.

**f. Documentation and Review:** The identified requirements are documented in this Software Requirements Specification (SRS) document. The document serves as a reference for system development, testing, review, and future enhancements, with updates made as requirements evolve.

### 3.1.2 Rationale for Selected Techniques

The requirements gathering techniques were selected to ensure that Vishvin ERP requirements align with organizational needs, existing business processes, and the intended functionality of the system.

**The above techniques were selected for the following reasons:**

1. **Management Guidance and Discussions**

- Help identify business objectives, operational needs, and expected system functionality.
- Ensure that ERP requirements align with management priorities.

**2. Current Process Review**

- Helps understand existing manual and hardcopy-based business processes.
- Identifies operational challenges and opportunities for digitization.

**3. HTML Screen Review**

- Helps identify screen-level functionality, data fields, user interactions, and expected system behavior.
- Ensures that documented requirements remain aligned with the developed interfaces.

**4. Workflow and Use Case Analysis**

- Helps define the expected workflows for the Super Admin, PR–PO, and Daily Reporting modules.
- Identifies relevant process steps, user responsibilities, and approval requirements.

**5. Requirement Analysis and Validation**

- Helps identify ambiguities, dependencies, and gaps in the documented requirements.
- Supports clarification and refinement based on management inputs and project scope.

**6. Requirement Documentation and Prioritization**

- Organizes requirements according to the current ERP modules and implementation priorities.
- Provides a structured reference for development, testing, review, and future enhancements.

#### 3.1.3 Formats Used

The following formats and documentation methods are used during the requirements gathering and documentation process to maintain consistency, clarity, and traceability of requirements for Vishvin ERP (VERP):

1. **Software Requirements Specification (SRS) Template**
    - Used to document the system scope, business requirements, functional requirements, and non-functional requirements in a structured format.
2. **Functional Requirements Matrix**
    - Tabular format listing functional features such as Company product,PO-PR Generation,Daily Work tracking, and reporting.
    - Mapped each feature to applicable user roles and system modules for traceability.
3. **Role & Access Control Matrix**
    - Used to define role-based permissions for Admin, Field Executive, Coordinator, and Site Executive roles.
    - Clearly specified allowed and restricted actions across Web and Mobile platforms.
4. **Workflow Diagrams**
    - Visual flow diagrams used to represent end-to-end workforce processes, including candidate lifecycle, onboarding workflow, attendance capture, and status transitions.
    - Helped identify control points, validations, and exception handling.
5. **UI Mock-ups and Feedback Forms**
    - Used to collect structured feedback on early UI concepts for Web and Mobile applications.
    - Focused on usability, clarity, and ease of execution for field users.
6. **Meeting Minutes (MoM) and Clarification Logs**
    - Formal templates used to record decisions, clarifications, action items, and approvals from requirement workshops and review meetings.
    - Ensured alignment and traceability of decisions throughout the requirements phase.

### 3.2 References to Customer's Documents - VERP Project

- **Existing HTML Screens:** Used to understand the current screen layouts, data fields, user interactions, and functionality of the Super Admin, PR–PO, and Daily Reporting modules.
- **Existing Business Records and Processes:** Manual records and hardcopy documentation, where available, serve as references for understanding current business operations and identifying digitization requirements.
- **Management Inputs and Clarifications:** Information and guidance provided by the management team are used to identify business requirements, expected functionality, and implementation priorities.
- **Standard SRS Template:** Used as the reference structure for documenting system requirements, constraints, scope, and other project-related information.

- **High-Level Architecture and Process Diagrams:** Used to represent the proposed system structure, module interactions, and high-level business workflows.
- **SRS Revision History:** Maintained to document changes to requirements, scope, and module coverage as Vishvin ERP evolves.

## 3.3 Functional Requirements

### 3.3.1 Login Module (A-Login)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation |
|---|---|---|---|---|---|
| Login — Username, Email or Phone | A-LOGIN-001 | High | The system shall allow registered users to enter their username, email address, or phone number or Vishvin's Employee ID (Length: 6) as the login identifier. | Login Screen UI | Verify login using valid registered identifiers and reject invalid or unregistered identifiers. |
| Login — Password | A-LOGIN-002 | High | The system shall allow users to enter their password securely. The password field shall provide an option to show or hide the entered password. | Login Screen UI | Verify password entry, password masking, show/hide functionality, and authentication with valid and invalid passwords. |
| Remember Me | A-LOGIN-003 | Medium | The system shall provide a Remember Me option to retain the user's login preference in accordance with the configured authentication and session policy. | Login Screen UI | Verify that the selected preference behaves according to the configured authentication policy. |
| Forgot Password | A-LOGIN-004 | High | The system shall provide a Forgot Password option to initiate the password recovery process for registered users. | Login Screen UI | Verify navigation to the password recovery workflow and handling of registered and unregistered identifiers. |
| Sign In to Dashboard | A-LOGIN-005 | High | The system shall authenticate the submitted credentials and allow authorized users to access the appropriate dashboard based on their assigned role and permissions. | Login Screen UI | Verify successful login, rejection of invalid credentials, and role-based redirection. |
| Google Sign-In | A-LOGIN-006 | Medium | The system shall provide an option to authenticate users through Google, subject to the availability and configuration of the required integration. | Login Screen UI | Verify successful and unsuccessful authentication scenarios when Google integration is enabled. |

| | | | | | |
|---|---|---|---|---|---|
| Facebook Sign-In | A-LOGIN-007 | Medium | The system shall provide an option to authenticate users through Facebook, subject to the availability and configuration of the required integration. | Login Screen UI | Verify successful and unsuccessful authentication scenarios when Facebook integration is enabled. |
| Sign Up | A-LOGIN-008 | Medium | The system shall provide a Sign Up option to direct users to the registration process, subject to the organization's user registration policy. | Login Screen UI | Verify navigation to the registration workflow and enforce the configured registration policy. |
| Theme Toggle | A-LOGIN-009 | Low | The system shall provide a theme toggle to allow users to switch between supported visual themes on the login screen. | Login Screen UI | Verify theme switching and the visual consistency of the login screen. |
| Login Screen Branding and Informatio n | A-LOGIN-010 | Medium | The system shall display the Vishvin ERP branding, Enterprise Cloud Portal identification, and designated informational content on the login screen. | Login Screen UI | Verify the logo, portal name, promotional content, layout, and display across supported screen sizes. |

### 3.3.A: Super Admin Module (Web Application)

### 3.3.A.1 : Company Creation (SA-COM-A)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Companies – View List | SA-COM-A-001 | High | The system shall display registered company records in a centralized table, including Company Code, Company Name, Email, Phone Number, Address, Status, and available Actions. | Companies Screen | Functional Testing, UAT | Master Management |
| Company Creation | SA-COM-A-002 | High | The system shall allow authorized users to initiate company creation using the Add New option. | Companies Screen | Functional Testing | Master Management |
| Company Details | SA-COM-A-003 | High | The system shall provide fields to enter Company Code, Company Name, Email, Phone Number, and Address when creating a company record. | Add New Company Form | Functional Testing, UAT | Master Management |

| | | | | | | |
|---|---|---|---|---|---|---|
| Save Company | SA-COM-A-004 | High | The system shall allow users to save the entered company details using the Save option. | Add New Company Form | Functional Testing, UAT | Master Management |
| Cancel Creation | SA-COM-A-005 | Medium | The system shall provide a Cancel option to exit the company creation process without intentionally saving the entered record. | Add New Company Form | Functional Testing | Master Management |
| Navigation | SA-COM-A-006 | Low | The system shall provide options to close the company creation form and return to the company listing. | Add New Company Form | Functional Testing | Navigation |
| Search Companies | SA-COM-A-007 | Medium | The system shall provide a search facility to locate company records using supported search criteria. | Companies Screen | Functional Testing | Master Management |
| Refresh List | SA-COM-A-008 | Low | The system shall provide an option to refresh the company listing. | Companies Screen | Functional Testing | Master Management |
| Column Configuration | SA-COM-A-009 | Low | The system shall provide an option to configure the columns displayed in the company listing. | Companies Screen | Functional Testing | Master Management |
| Export Companies | SA-COM-A-010 | Medium | The system shall provide an option to export company records using the available export functionality. | Companies Screen | Functional Testing | Master Management |

### 3.3.A.2 : Product Creation (SA-PROD-B)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Products – View List | SA-PROD-B-001 | High | The system shall display registered products in a centralized table with Product Code, Product Name, Description, Status, and available Actions. | Products Screen | Functional Testing, UAT | Master Management |

| | | | | | | |
|---|---|---|---|---|---|---|
| Product Creation | SA-PROD-B-002 | High | The system shall allow authorized users to initiate product creation using the Add New option. | Products Screen | Functional Testing | Master Management |
| Product Details | SA-PROD-B--003 | High | The system shall provide fields to enter Product Code, Product Name, and Description when creating a product record. | Add New Product Form | Functional Testing, UAT | Master Management |
| Save Product | SA-PROD-B--004 | High | The system shall allow users to save the entered product details using the Save option. | Add New Product Form | Functional Testing, UAT | Master Management |
| Cancel Creation | SA-PROD-B--005 | Medium | The system shall provide a Cancel option to exit the product creation process without saving the new record. | Add New Product Form | Functional Testing | Master Management |
| Navigation | SA-PROD-B--006 | Low | The system shall provide options to close the product creation form and return to the product listing. | Add New Product Form | Functional Testing | Navigation |
| Search Products | SA-PROD-B--007 | Medium | The system shall provide a search facility to locate product records using supported search criteria. | Products Screen | Functional Testing | Master Management |
| Refresh List | SA-PROD-B--008 | Low | The system shall provide an option to refresh the product listing. | Products Screen | Functional Testing | Master Management |
| Column Configuration | SA-PROD-B--009 | Low | The system shall provide an option to configure the columns displayed in the product listing. | Products Screen | Functional Testing | Master Management |
| Export Products | SA-PROD-B--010 | Medium | The system shall provide an option to export product records using the available export functionality. | Products Screen | Functional Testing | Master Management |

### 3.3.A.C : Company Product Creation (SA-COPR-C)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference SA-COPR-C | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Company Products – View List | SA-COPR-C 001 | High | The system shall display company-product mappings in a centralized table, including Company Name, Product Name, Created On, Status, and available Actions. | Company Products Screen | Functional Testing, UAT | Master Management |
| Company Product Creation | SA-COPR-C-002 | High | The system shall allow authorized users to initiate a new company-product mapping using the Add New option. | Company Products Screen | Functional Testing | Master Management |
| Company and Product Selection | SA-COPR-C-003 | High | The system shall provide dropdown fields to select an existing Company Name and Product Name when creating a company-product mapping. | Add New Company Product Form | Functional Testing, UAT | Master Management |
| Save Company Product | SA-COPR-C-004 | High | The system shall allow users to save the selected company-product mapping using the Save option. | Add New Company Product Form | Functional Testing, UAT | Master Management |
| Cancel Creation | SA-COPR-C-005 | Medium | The system shall provide a Cancel option to exit the creation process without saving the new mapping. | Add New Company Product Form | Functional Testing | Master Management |
| Navigation | SA-COPR-C-006 | Low | The system shall provide options to close the creation form and return to the company-product listing. | Add New Company Product Form | Functional Testing | Navigation |
| Search Company Products | SA-COPR-C-007 | Medium | The system shall provide a search facility to locate company-product mappings using supported search criteria. | Company Products Screen | Functional Testing | Master Management |
| Refresh List | SA-COPR-C-008 | Low | The system shall provide an option to refresh the company-product listing. | Company Products Screen | Functional Testing | Master Management |

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| Column Configuration | SA-COPR-C-009 | Low | The system shall provide an option to configure the columns displayed in the listing. | Company Products Screen | Functional Testing | Master Management |
| Export Company Products | SA-COPR-C-010 | Medium | The system shall provide an option to export company-product mapping records using the available export functionality. | Company Products Screen | Functional Testing | Master Management |
| Mapping Status Management | SA-COPR-C-011 | Medium | The system shall provide controls to manage the status of a company-product mapping, subject to the available actions and authorized permissions. | Company Products Screen | Functional Testing, UAT | Master Management |
| Mapping Record Actions | SA-COPR-C-012 | Medium | The system shall provide available record-level actions, including viewing, editing, and deleting company-product mappings, as displayed on the screen. | Company Products Screen | Functional Testing, UAT | Master Management |

### 3.3.A.D : Projects Creation (SA-PROJ-D)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference SA-PROJ-D | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Projects – View List | SA-PROJ-D-001 | High | The system shall display project records in a centralized table, including Project Code, Project Name, Company, Product, Description, Status, and available Actions. | Projects Screen | Functional Testing, UAT | Master Management |
| Project Creation | SA-PROJ-D-002 | High | The system shall allow authorized users to initiate project creation using the Add New option. | Projects Screen | Functional Testing | Master Management |
| Company Product Selection | SA-PROJ-D-003 | High | The system shall provide a dropdown to select an existing Company Product when creating a project. | Add New Project Form | Functional Testing, UAT | Master Management |
| Project Details | SA-PROJ-D-004 | High | The system shall provide fields to enter Project Code, Project Name, and Description. | Add New Project Form | Functional Testing, UAT | Master Management |

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| Save Project | SA-PROJ-D-005 | High | The system shall allow users to save the entered project details using the Save option. | Add New Project Form | Functional Testing, UAT | Master Management |
| Cancel Creation | SA-PROJ-D-006 | Medium | The system shall provide a Cancel option to exit project creation without saving the new record. | Add New Project Form | Functional Testing | Master Management |
| Navigation | SA-PROJ-D-007 | Low | The system shall provide options to close the project creation form and return to the project listing. | Add New Project Form | Functional Testing | Navigation |
| Search Projects | SA-PROJ-D-008 | Medium | The system shall provide a search facility to locate project records using supported search criteria. | Projects Screen | Functional Testing | Master Management |
| Refresh List | SA-PROJ-D-009 | Low | The system shall provide an option to refresh the project listing. | Projects Screen | Functional Testing | Master Management |
| Column Configuration | SA-PROJ-D-010 | Low | The system shall provide an option to configure the columns displayed in the project listing. | Projects Screen | Functional Testing | Master Management |
| Export Projects | SA-PROJ-D-011 | Medium | The system shall provide an option to export project records using the available export functionality. | Projects Screen | Functional Testing | Master Management |
| Project Status Management | SA-PROJ-D-012 | Medium | The system shall provide controls to manage project status according to available actions and authorized permissions. | Projects Screen | Functional Testing, UAT | Master Management |
| Project Record Actions | SA-PROJ-D-013 | Medium | The system shall provide available record-level actions, including viewing, editing, and deleting project records, as displayed on the screen. | Projects Screen | Functional Testing, UAT | Master Management |

### 3.3.A.E : Designation Creation (SA-DES-E)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference SA-DES-E | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Designations – View List | SA-DES-E-001 | High | The system shall display designation records in a centralized table, including Designation Code, Designation Name, Company, Status, and available Actions. | Designations Screen | Functional Testing, UAT | Master Management |
| Designation Creation | SA-DES-E-002 | High | The system shall allow authorized users to initiate designation creation using the Add New option. | Designations Screen | Functional Testing | Master Management |
| Company Selection | SA-DES-E-003 | High | The system shall provide a dropdown to select an existing company when creating a designation. | Add New Designation Form | Functional Testing, UAT | Master Management |
| Designation Details | SA-DES-E-004 | High | The system shall provide fields to enter Designation Code, Designation Name, and Designation Level. | Add New Designation Form | Functional Testing, UAT | Master Management |
| Save Designation | SA-DES-E-005 | High | The system shall allow users to save the entered designation details using the Save option. | Add New Designation Form | Functional Testing, UAT | Master Management |
| Cancel Creation | SA-DES-E-006 | Medium | The system shall provide a Cancel option to exit designation creation without saving the new record. | Add New Designation Form | Functional Testing | Master Management |
| Navigation | SA-DES-E-007 | Low | The system shall provide options to close the designation creation form and return to the designation listing. | Add New Designation Form | Functional Testing | Navigation |
| Search Designations | SA-DES-E-008 | Medium | The system shall provide a search facility to locate designation records using supported search criteria. | Designations Screen | Functional Testing | Master Management |
| Refresh List | SA-DES-E-009 | Low | The system shall provide an option to refresh the designation listing. | Designations Screen | Functional Testing | Master Management |

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| Column Configuration | SA-DES-E-010 | Low | The system shall provide an option to configure the columns displayed in the designation listing. | Designations Screen | Functional Testing | Master Management |
| Export Designations | SA-DES-E-011 | Medium | The system shall provide an option to export designation records using the available export functionality. | Designations Screen | Functional Testing | Master Management |
| Designation Status Management | SA-DES-E-012 | Medium | The system shall provide controls to manage designation status according to available actions and authorized permissions. | Designations Screen | Functional Testing, UAT | Master Management |
| Designation Record Actions | SA-DES-E-013 | Medium | The system shall provide available record-level actions, including viewing, editing, and deleting designation records, as displayed on the screen. | Designations Screen | Functional Testing, UAT | Master Management |

### 3.3.A.F : Roles & Permission Creation (SA-RP-F)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Roles – View List | SA-RP-F-001 | High | The system shall display registered roles with Role Code, Role Name, Company, Description, Status, and available Actions. | Roles Screen | Functional Testing, UAT | Master Management |
| Role Creation | SA-RP-F-002 | High | The system shall allow authorized users to initiate role creation using the Add New option. | Roles Screen | Functional Testing | Master Management |
| Company Selection | SA-RP-F-003 | High | The system shall provide a dropdown to select an existing company when creating a role. | Add New Role Form | Functional Testing, UAT | Master Management |
| Role Details | SA-RP-F-004 | High | The system shall provide fields to enter Role Code, Role Name, and Description. | Add New Role Form | Functional Testing, UAT | Master Management |

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| Save Role | SA-RP-F-005 | High | The system shall allow authorized users to save the entered role details using the Save option. | Add New Role Form | Functional Testing, UAT | Master Management |
| Cancel Creation | SA-RP-F-006 | Medium | The system shall provide a Cancel option to exit role creation without saving the new record. | Add New Role Form | Functional Testing | Master Management |
| Navigation | SA-RP-F-007 | Low | The system shall provide options to close the role creation form and return to the role listing. | Add New Role Form | Functional Testing | Navigation |
| Search Roles | SA-RP-F-008 | Medium | The system shall provide a search facility to locate role records using supported search criteria. | Roles Screen | Functional Testing | Master Management |
| Refresh List | SA-RP-F-009 | Low | The system shall provide an option to refresh the role listing. | Roles Screen | Functional Testing | Master Management |
| Column Configuration | SA-RP-F-010 | Low | The system shall provide an option to configure the columns displayed in the role listing. | Roles Screen | Functional Testing | Master Management |
| Export Roles | SA-RP-F-011 | Medium | The system shall provide an option to export role records using the available export functionality. | Roles Screen | Functional Testing | Master Management |
| Role Status Management | SA-RP-F-012 | Medium | The system shall provide controls to manage role status according to available actions and authorized permissions. | Roles Screen | Functional Testing, UAT | Master Management |
| Role Record Actions | SA-RP-F-013 | Medium | The system shall provide available record-level actions, including viewing, editing, and deleting role records, as displayed on the screen. | Roles Screen | Functional Testing, UAT | Master Management |

### 3.3.A.G : User Creation (SA-UC-G)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| User Creation Wizard | SA-UC-G-001 | High | The system shall provide a step-by-step wizard to register a new system user, covering Personal Information, Account Credentials, Company Assignment, Profile Photo, and Supporting Documents. | User Creation Wizard | Functional Testing, UAT | User Management |
| Personal Information | SA-UC-G-002 | High | The system shall provide fields to capture Employee ID, First Name, Middle Name, Last Name, Phone Number, Alternate Phone, Date of Birth, Date of Joining, Gender, and Marital Status. | Step 1 – Personal Information | Functional Testing, UAT | User Profile |
| Mandatory Field Validation | SA-UC-G-003 | High | The system shall validate mandatory personal information fields before allowing the user to proceed to the next step. | Step 1 – Personal Information | Validation Testing | User Profile |
| Account Credentials | SA-UC-G-004 | High | The system shall provide fields to configure Email Address, Login Name, and Password for the new user. | Step 2 – Account Credentials | Functional Testing, UAT | Authentication |
| Credential Validation | SA-UC-G-005 | High | The system shall validate mandatory credential fields and enforce applicable login and password validation rules before proceeding. | Step 2 – Account Credentials | Validation Testing | Authentication |
| Company Assignment | SA-UC-G-006 | High | The system shall allow the administrator to assign the user to a Company, Company Product, Project, Role, and Designation, with Reporting Manager as an optional assignment. | Step 3 – Company Assignment | Functional Testing, UAT | User Assignment |
| Dependent Dropdowns | SA-UC-G-007 | High | The system shall provide relevant selection options for Company Product, Project, Role, Designation, and Reporting Manager based on the available records and applicable dependencies. | Step 3 – Company Assignment | Integration Testing, UAT | User Assignment |
| Profile Photo Upload | SA-UC-G-008 | Medium | The system shall allow the administrator to upload or change the user's profile photo in supported formats (JPG, PNG, and WEBP), with a maximum file size of 5 MB. | Step 4 – Profile Photo | Functional Testing, File Validation | User Profile |

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| Supporting Documents | SA-UC-G-009 | Medium | The system shall provide separate upload options for Aadhaar Card, PAN Card, Other1, and Others2 files(This Upload Document names will be finalized shortly). The displayed form identifies these document fields as optional. | Step 5 – Supporting Documents | Functional Testing, File Validation | Document Management |
| User Summary | SA-UC-G-010 | Medium | The system shall display a summary of the employee, login username, company, and profile photo upload status before final user creation. | Step 5 – Supporting Documents | Functional Testing, UAT | User Management |
| Step Navigation | SA-UC-G-011 | Medium | The system shall provide Next Step and Previous navigation options to move between wizard steps. | User Creation Wizard | Functional Testing | Navigation |
| Reset Form | SA-UC-G-012 | Medium | The system shall provide a Reset Form option to clear or reset the user creation form. | User Creation Wizard | Functional Testing | Form Management |
| Create User | SA-UC-G-013 | High | The system shall allow the administrator to initiate user creation using the Create User option after completing the applicable required fields and validations. | Step 5 – Supporting Documents | Functional Testing, UAT | User Management |

### 3.3.A.H : Manage Users (SA-MU-H)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Manage Users – View List | SA-MU-H-001 | High | The system shall display registered user records, including Employee ID, User Name, Email, Company, Role Name, Reporting Manager, Created At, Status, and available Actions. | Manage Users Screen | Functional Testing, UAT | User Management |
| Search Users | SA-MU-H-002 | Medium | The system shall provide a search facility to locate user records using supported search criteria. | Manage Users Screen | Functional Testing | User Management |

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|---|---|---|---|---|---|---|
| Refresh User List | SA-MU-H-003 | Low | The system shall provide an option to refresh the user listing. | Manage Users Screen | Functional Testing | User Management |
| Column Configuration | SA-MU-H-004 | Low | The system shall provide an option to configure the columns displayed in the user listing. | Manage Users Screen | Functional Testing | User Management |
| User Status Management | SA-MU-H-005 | High | The system shall provide a control to activate or deactivate a user account, subject to authorized permissions. | Manage Users Screen | Functional Testing, UAT | User Management |
| View User Details | SA-MU-H-006 | Medium | The system shall provide an option to view the details of an existing user. | Manage Users Screen | Functional Testing, UAT | User Management |
| Edit User Details | SA-MU-H-007 | High | The system shall provide an option to edit an existing user's details through the available edit functionality. | Manage Users Screen | Functional Testing, UAT | User Management |
| User Record Identification | SA-MU-H-008 | Medium | The system shall display the user's Employee ID and associated profile information to support identification of individual user records. | Manage Users Screen | Functional Testing | User Management |

### 3.3.A.H : Manage Page (SA-MP-I)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Manage Pages – View List | SA-MP-I-001 | High | The system shall display configured application pages with Page Name, Icon Name, Page URL, Parent Page, Order No, Role Name, Status, and available Actions. | Manage Pages Screen | Functional Testing, UAT | Page Management |
| Search Pages | SA-MP-I-002 | Medium | The system shall provide a search facility to locate configured pages using supported search criteria. | Manage Pages Screen | Functional Testing | Page Management |

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|---|---|---|---|---|---|---|
| Refresh Page List | SA-MP-I-003 | Low | The system shall provide an option to refresh the page listing. | Manage Pages Screen | Functional Testing | Page Management |
| Column Configuration | SA-MP-I-004 | Low | The system shall provide an option to configure the columns displayed in the page listing. | Manage Pages Screen | Functional Testing | Page Management |
| Add New Page | SA-MP-I-005 | High | The system shall allow the Super Admin to create a new application page by entering the mandatory Page Name, Page URL, Icon Name, Page Type, and Role Name. | Add New Page Form | Functional Testing, UAT | Page Configuration |
| Page Type Selection | SA-MP-I-006 | High | The system shall allow the Super Admin to classify a page as Main Page, Parent Page, or Child Page. | Add New Page Form | Functional Testing | Page Hierarchy |
| Page Order Configuration | SA-MP-I-007 | Medium | The system shall provide an option to configure the display order of application pages when required. | Add New Page Form | Functional Testing | Page Hierarchy |
| Role Assignment | SA-MP-I-008 | High | The system shall allow the Super Admin to assign one or more applicable roles to a page, subject to the supported role assignment configuration. | Add New Page Form | Functional Testing, UAT | Access Control |
| Save Page | SA-MP-I-009 | High | The system shall validate mandatory fields and save the page configuration when the required information is provided correctly. | Add New Page Form | Functional Testing, UAT | Page Configuration |

### 3.3.A.H : Manage Categories (SA-MC-J)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Manage Pages – View List | SA-MC-I-001 | High | The system shall display configured application pages with Page Name, Icon Name, Page URL, Parent Page, Order No, Role Name, Status, and available Actions. | Manage Pages Screen | Functional Testing, UAT | Page Management |

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|---|---|---|---|---|---|---|
| Search Pages | SA-MC-I-002 | Medium | The system shall provide a search facility to locate configured pages using supported search criteria. | Manage Pages Screen | Functional Testing | Page Management |
| Refresh Page List | SA-MC-I-003 | Low | The system shall provide an option to refresh the page listing. | Manage Pages Screen | Functional Testing | Page Management |
| Column Configuration | SA-MC-I-004 | Low | The system shall provide an option to configure the columns displayed in the page listing. | Manage Pages Screen | Functional Testing | Page Management |
| Add New Page | SA-MC-I-005 | High | The system shall allow the Super Admin to create a new application page by entering the mandatory Page Name, Page URL, Icon Name, Page Type, and Role Name. | Add New Page Form | Functional Testing, UAT | Page Configuration |
| Page Type Selection | SA-MC-I-006 | High | The system shall allow the Super Admin to classify a page as Main Page, Parent Page, or Child Page. | Add New Page Form | Functional Testing | Page Hierarchy |
| Page Order Configuration | SA-MC-I-007 | Medium | The system shall provide an option to configure the display order of application pages when required. | Add New Page Form | Functional Testing | Page Hierarchy |
| Role Assignment | SA-MC-I-008 | High | The system shall allow the Super Admin to assign one or more applicable roles to a page, subject to the supported role assignment configuration. | Add New Page Form | Functional Testing, UAT | Access Control |
| Save Page | SA-MC-I-009 | High | The system shall validate mandatory fields and save the page configuration when the required information is provided correctly. | Add New Page Form | Functional Testing, UAT | Page Configuration |

### 3.3.A.H : Material Creation (SA-MTC-K)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
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| Item Creation Form | SA-MTC-K-001 | High | The system shall provide a form to create procurement items by capturing item details, classification, unit of measurement, tax information, and other applicable attributes. | Item Creation Screen | Functional Testing, UAT | Item Management |
| Item Name | SA-MTC-K-002 | High | The system shall require the user to enter the item name before saving an item. | Item Creation Form | Field Validation Testing | Item Management |
| Item Code | SA-MTC-K-003 | High | The system shall require a unique item code for each item created in the system. | Item Creation Form | Field Validation, Duplicate Check | Item Management |
| Category Selection | SA-MTC-K-004 | High | The system shall allow the user to select an item category from the configured item categories. | Item Creation Form | Functional Testing | Item Classification |
| Subcategory Selection | SA-MTC-K-005 | High | The system shall allow the user to select an item subcategory from the configured subcategories. | Item Creation Form | Functional Testing | Item Classification |
| Item Attributes | SA-MTC-K-006 | Medium | The system shall allow the user to enter optional item attributes, including Brand and Model Number. | Item Creation Form | Functional Testing | Item Details |
| Unit of Measurement | SA-MTC-K-007 | High | The system shall require the user to select the applicable Unit of Measurement (UOM) for the item. | Item Creation Form | Field Validation Testing | Item Details |
| Tax and Warranty Details | SA-MTC-K-008 | Medium | The system shall support entry of HSN/SAC code, GST percentage, and warranty period, where applicable. | Item Creation Form | Functional Testing | Item Details |
| Specification, Description and Remarks | SA-MTC-K-009 | Medium | The system shall allow the user to enter item specifications, description, and remarks. | Item Creation Form | Functional Testing | Item Details |

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| Item Status | SA-MTC-K-010 | High | The system shall allow the user to set the item status as Active or other configured status values. | Item Creation Form | Functional Testing | Item Management |
| Save Item | SA-MTC-K-011 | High | The system shall validate mandatory fields and save the item when the entered information is valid. | Item Creation Form | Functional Testing, UAT | Item Management |
| Clear Form | SA-MTC-K-012 | Low | The system shall provide a Clear option to reset the item creation form. | Item Creation Form | Functional Testing | Item Management |
| Items List | SA-MTC-K-013 | High | The system shall display the list of created items with relevant details, including item code, name, category, subcategory, brand, model number, UOM, HSN/SAC, GST, warranty, and status. | Items List | Functional Testing, UAT | Item Management |
| Search Items | SA-MTC-K-014 | Medium | The system shall provide a search facility to locate items in the Items List. | Items List | Functional Testing | Item Management |
| Edit Item | SA-MTC-K-015 | High | The system shall allow authorized users to edit the details of an existing item. | Items List | Functional Testing, UAT | Item Management |
| Delete Item | SA-MTC-K-016 | Medium | The system shall provide an option to delete an existing item, subject to applicable authorization and usage restrictions. | Items List | Functional Testing | Item Management |

### 3.3.A.H : Material Sub Category (SA-MSC-L)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Subcategory Creation | SA-MSC-L-001 | High | The system shall provide a form to create a material subcategory by selecting a material category and entering the subcategory code and name. | Create Sub Category Screen | Functional Testing, UAT | Material Classification |

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| Material Category Selection | SA-MSC-L-002 | High | The system shall require the user to select a material category from the configured categories. | Create Sub Category Form | Field Validation Testing | Material Classification |
| Subcategory Code | SA-MSC-L-003 | High | The system shall require a unique subcategory code for each material subcategory. | Create Sub Category Form | Field Validation, Duplicate Check | Material Classification |
| Subcategory Name | SA-MSC-L-004 | High | The system shall require the user to enter a subcategory name before saving. | Create Sub Category Form | Field Validation Testing | Material Classification |
| Create Subcategory | SA-MSC-L-005 | High | The system shall validate mandatory fields and save the subcategory when the entered information is valid. | Create Sub Category Form | Functional Testing, UAT | Material Classification |
| Subcategory List | SA-MSC-L-006 | High | The system shall display the material category, subcategory code, subcategory name, status, creation date, and creator in the subcategory list. | Manage Sub Categories Screen | Functional Testing, UAT | Material Classification |
| Search Subcategories | SA-MSC-L-007 | Medium | The system shall allow users to search subcategories by name or code. | Manage Sub Categories Screen | Functional Testing | Material Classification |
| Subcategory Status | SA-MSC-L-008 | High | The system shall allow authorized users to activate or deactivate a material subcategory. | Manage Sub Categories Screen | Functional Testing | Material Classification |
| View Subcategory | SA-MSC-L-009 | Medium | The system shall allow users with appropriate permissions to view the details of a material subcategory. | Manage Sub Categories Screen | Functional Testing | Material Classification |
| Edit Subcategory | SA-MSC-L-010 | High | The system shall allow authorized users to update existing subcategory details, subject to applicable validation rules. | Manage Sub Categories Screen | Functional Testing, UAT | Material Classification |

### 3.3.B: PO-PR Module (POPR)

### 3.3.B.1 : Dashboard (POPR-DASH-A)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Procurement Dashboard | POPR-DASH-A-001 | High | The system shall display a procurement dashboard providing an overview of procurement operations. | Dashboard UI | Verify that the dashboard loads and displays the available procurement information. | POPR |
| Total PRs | POPR-DASH-A-002 | High | The system shall display the total number of Purchase Requisitions (PRs). | Dashboard KPI | Verify that the displayed count matches the underlying PR records. | POPR |
| Pending Approval | POPR-DASH-A-003 | High | The system shall display the number of Purchase Requisitions awaiting approval. | Dashboard KPI | Verify that the count reflects PRs currently pending approval. | POPR |
| Approved PRs | POPR-DASH-A-004 | High | The system shall display the number of approved Purchase Requisitions. | Dashboard KPI | Verify that the count matches approved PR records. | POPR |
| Generated POs | POPR-DASH-A-005 | High | The system shall display the number of Purchase Orders generated from the procurement process. | Dashboard KPI | Verify that the count matches generated PO records. | POPR |
| Total PO Value | POPR-DASH-A-006 | High | The system shall display the aggregate value of generated Purchase Orders. | Dashboard KPI | Verify that the displayed value matches the total value of applicable PO records. | POPR |

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| Active Vendors | POPR-DASH-A-007 | Medium | The system shall display the number of active vendors registered in the system. | Dashboard KPI | Verify that the count matches vendors classified as active. | POPR |
| Documents Uploaded | POPR-DASH-A-008 | Medium | The system shall display the number of documents uploaded and tracked in the procurement process. | Dashboard KPI | Verify that the count matches the applicable uploaded document records. | POPR |
| Pending Payments | POPR-DASH-A-009 | High | The system shall display the aggregate value of payments pending release. | Dashboard KPI | Verify that the displayed amount matches the applicable pending payment records. | POPR |
| Export Report | POPR-DASH-A-010 | Medium | The system shall provide an option to export the procurement dashboard report. | Export Report button | Verify that the export action generates a report containing the supported dashboard information. | POPR |
| Monthly PO Value Trend | POPR-DASH-A-011 | Medium | The system shall display a monthly trend chart showing Purchase Order values over the displayed period. | Monthly PO Value Trend chart | Verify that monthly values correspond to the underlying PO records and displayed period. | POPR |
| PR Status Distribution | POPR-DASH-A-012 | Medium | The system shall display the distribution of Purchase Requisitions by status, including the categories represented in the dashboard legend. | PR Status Distribution chart | Verify that the chart reflects the correct PR counts for each status category. | POPR |

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|---|---|---|---|---|---|---|
| Recent Activity | POPR-DASH-A-013 | Medium | The system shall display recent procurement activities, including the associated user, activity description, and timestamp where available. | Recent Activity section | Verify that displayed activities correspond to recorded system events and their timestamps. | POPR |
| User Information | POPR-DASH-A-014 | Low | The system shall display the logged-in user's name, role, and profile indicator. | Dashboard header | Verify that the displayed user information corresponds to the authenticated account. | POPR |
| Notifications | POPR-DASH-A-015 | Low | The system shall display a notification indicator showing the number of available notifications. | Notification icon | Verify that the notification count corresponds to the user's applicable notifications. | POPR |

### 3.3.B.2: Raise PR's Request (POPR-RPRR-B)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Raise PR's Request – Screen | POPR-RPRR-B-001 | High | The system shall provide a Purchase Requisition Form to allow authorized users to raise a new purchase requisition request. | Raise PR's Request Screen | Functional Testing, UAT | Purchase Requisition Management |
| Requirement Details | POPR-RPRR-B-002 | High | The system shall display the Requirement No, Request Date, and Requested By fields to identify the purchase requisition request. | Purchase Requisition Form | Functional Testing, UAT | Request Information |
| Employee Information | POPR-RPRR-B-003 | Medium | The system shall display the requester's Designation, Department, and Reporting To details in the purchase requisition form. | Purchase Requisition Form | Functional Testing | Request Information |
| Project Selection | POPR-RPRR-B-004 | High | The system shall provide a Project dropdown to allow the requester to select the project for which the purchase requisition is being raised. | Purchase Requisition Form | Functional Testing, Validation Testing | Request Details |
| Priority Selection | POPR-RPRR-B-005 | High | The system shall provide a mandatory Priority dropdown to allow the requester to | Purchase Requisition Form | Functional Testing, Validation Testing | Request Details |

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| | | | specify the priority of the purchase requisition. | | | |
| Required Date | POPR-RPRR-B-006 | High | The system shall provide a mandatory Required Date field to specify the date by which the requested items are needed. | Purchase Requisition Form | Functional Testing, Validation Testing | Request Details |
| Purpose / Justification | POPR-RPRR-B-007 | High | The system shall provide a Purpose / Justification field to allow the requester to enter the reason for raising the purchase requisition. | Purchase Requisition Form | Functional Testing, UAT | Request Details |
| Item Selection | POPR-RPRR-B-008 | High | The system shall provide an Item Name dropdown for each item row to allow the requester to select items from the available item records. | Items Section | Functional Testing, Integration Testing | Item Management |
| Item Quantity | POPR-RPRR-B-009 | High | The system shall provide a Qty field for each selected item to allow the requester to specify the required quantity. | Items Section | Functional Testing, Validation Testing | Item Management |
| Add Item | POPR-RPRR-B-010 | High | The system shall provide an Add Item option to allow the requester to add additional item rows to the purchase requisition form. | Items Section | Functional Testing, UAT | Item Management |
| Remove Item | POPR-RPRR-B-011 | Medium | The system shall provide a Remove option for each item row to allow the requester to remove an item from the purchase requisition before saving or submitting the request. | Items Section | Functional Testing | Item Management |
| Item Serial Number | POPR-RPRR-B-012 | Low | The system shall display a serial number for each item row to distinguish individual items listed in the purchase requisition. | Items Section | Functional Testing | Item Management |
| Attachments | POPR-RPRR-B-013 | Medium | The system shall provide an attachment upload option to allow the requester to attach supporting files to the purchase requisition. | Attachments Section | Functional Testing, File Validation | Document Management |
| Save Draft | POPR-RPRR-B-014 | High | The system shall provide a Save Draft option to allow the requester to save the purchase requisition without submitting it for further processing. | Purchase Requisition Form | Functional Testing, Data Persistence Testing | Request Management |
| Submit Requirement | POPR-RPRR-B-015 | High | The system shall provide a Submit Requirement option to submit the purchase requisition for the applicable approval workflow after the required information and validations are completed. | Purchase Requisition Form | Functional Testing, Validation Testing, UAT | Request Management |
| Clear Form | POPR-RPRR-B-016 | Medium | The system shall provide a Clear option to clear the entered purchase requisition details from the form. | Purchase Requisition Form | Functional Testing | Form Management |

### 3.3.B: View Raise PR's (POPR-VRPR-C)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| View Raise PR's – View List | POPR-VR PR-C-001 | High | The system shall display all purchase requisitions raised by the logged-in user in a centralized table. | View Raise PR's Screen | Functional Testing, UAT | Purchase Requisition Management |
| Screen Description | POPR-VR PR-C-002 | Low | The system shall display a description indicating that users can view all their raised purchase requisitions. | View Raise PR's Screen | Functional Testing | Navigation |
| Purchase Requisition Listing | POPR-VR PR-C-003 | High | The system shall display purchase requisition records with Sl No, Requirement No, Project, Priority, Required Date, Requested By, Status, and Actions. | My Purchase Requisitions | Functional Testing, UAT | Request Management |
| Search Purchase Requisitions | POPR-VR PR-C-004 | Medium | The system shall provide a search field to allow users to search for purchase requisitions using supported search criteria. | My Purchase Requisitions | Functional Testing, Validation Testing | Search and Filtering |
| Record Count | POPR-VR PR-C-005 | Low | The system shall display the total number of purchase requisition records shown in the listing. | My Purchase Requisitions | Functional Testing | Request Management |
| View Requisition Details | POPR-VR PR-C-006 | High | The system shall provide a View action for each purchase requisition to allow users to open and review the selected requisition details. | My Purchase Requisitions | Functional Testing, UAT | Request Management |
| Requisition Identification | POPR-VR PR-C-007 | High | The system shall display the Requirement No, Request Date, and Requested By details for the selected purchase requisition. | Purchase Requisition Details | Functional Testing | Request Information |
| Requester Information | POPR-VR PR-C-008 | Medium | The system shall display the requester's Designation, Department, and Reporting To details for the selected purchase requisition. | Purchase Requisition Details | Functional Testing | Request Information |
| Requisition Details | POPR-VR PR-C-009 | High | The system shall display the selected requisition's Project, Priority, Required Date, and Purpose / Justification. | Purchase Requisition Details | Functional Testing, UAT | Request Information |

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| Requested Items | POPR-VR PR-C-010 | High | The system shall display the items associated with the selected purchase requisition, including the serial number, Item Name, and Qty. | Purchase Requisition Details – Items | Functional Testing, Data Validation | Item Management |
| Quotation Details | POPR-VR PR-C-011 | High | The system shall display the quotations associated with the selected purchase requisition, including the vendor name, quotation reference number, quotation attachment, and available quotation amount details. | Purchase Requisition Details – Quotations | Functional Testing, Integration Testing | Quotation Management |
| Supporting Attachments | POPR-VR PR-C-012 | Medium | The system shall display the supporting attachments associated with the selected purchase requisition and provide access to the available files. | Purchase Requisition Details – Attachments | Functional Testing, File Validation | Document Management |
| Approval History | POPR-VR PR-C-013 | High | The system shall display the approval history of the selected purchase requisition, including the user involved, action or status description, and corresponding date and time. | Purchase Requisition Details – Approval History | Functional Testing, UAT | Approval Management |
| Requisition Status | POPR-VR PR-C-014 | High | The system shall display the current status of each purchase requisition in the listing, such as Approved, as applicable to the record. | My Purchase Requisitions | Functional Testing, Data Validation | Request Management |
| Close Details View | POPR-VR PR-C-015 | Medium | The system shall provide a close option in the purchase requisition details view to allow users to close the displayed details and return to the previous screen. | Purchase Requisition Details | Functional Testing | Navigation |

### 3.3.B: Pending Approvals (POPR-MPA-D)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| My Pending Approvals – View List | POPR-MPA-D-001 | High | The system shall provide a My Pending Approvals screen to allow authorized users to review and take action on purchase requisitions pending their approval. | My Pending Approvals Screen | Functional Testing, UAT | Purchase Requisition Approval |
| Screen Description | POPR-MPA-D-002 | Low | The system shall display a description indicating that users can review and take | My Pending Approvals Screen | Functional Testing | Navigation |

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| | | | action on purchase requisitions pending approval. | | | |
| Pending Approval Listing | POPR-MP A-D-003 | High | The system shall display pending purchase requisitions in a centralised table containing SL No, Requirement No, Employee, Project, Priority, Required Date, and Action columns. | Pending Approval List | Functional Testing, UAT | Approval Management |
| Search Pending PRs | POPR-MP A-D-004 | Medium | The system shall provide a search field to allow authorised users to search pending purchase requisitions using supported search criteria. | Pending Approval List | Functional Testing, Validation Testing | Search and Filtering |
| Record Count | POPR-MP A-D-005 | Low | The system shall display the total number of pending approval records in the listing. | Pending Approval List | Functional Testing | Approval Management |
| Employee Details | POPR-MP A-D-006 | Medium | The system shall display the employee associated with each pending purchase requisition to help the approver identify the requester. | Pending Approval List | Functional Testing, Data Validation | Approval Management |
| Purchase Requisition Identification | POPR-MP A-D-007 | High | The system shall display the Requirement No for each pending purchase requisition to identify the request requiring approval. | Pending Approval List | Functional Testing | Approval Management |
| Project and Priority Details | POPR-MP A-D-008 | High | The system shall display the Project and Priority associated with each pending purchase requisition. | Pending Approval List | Functional Testing, Data Validation | Approval Management |
| Required Date | POPR-MP A-D-009 | High | The system shall display the Required Date for each pending purchase requisition to indicate when the requested items are needed. | Pending Approval List | Functional Testing, Data Validation | Approval Management |
| Pending Approval Action | POPR-MP A-D-010 | High | The system shall provide an Action option for each pending purchase requisition to allow the authorised approver to access the applicable approval operations. | Pending Approval List | Functional Testing, UAT | Approval Workflow |
| Empty State | POPR-MP A-D-011 | Medium | The system shall display an appropriate message, such as “No pending approvals,” when no purchase requisitions are awaiting approval. | Pending Approval List | Functional Testing, UAT | Approval Management |

### 3.3.B: Quotation Tracking (POPR-QT-E)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Quotation Tracking – Screen | POPR-QT-E-001 | High | The system shall provide a Quotation Tracking screen to allow authorized users to upload and manage vendor quotations associated with purchase requisitions. | Quotation Tracking Screen | Functional Testing, UAT | Quotation Management |
| Screen Description | POPR-QT-E-002 | Low | The system shall display a description indicating that users can upload and manage vendor quotations for purchase requisitions. | Quotation Tracking Screen | Functional Testing | Navigation |
| PR Number | POPR-QT-E-003 | High | The system shall provide a PR No. field to display the purchase requisition number associated with the selected requisition. | Quotation Tracking Screen | Functional Testing, Data Validation | Purchase Requisition Tracking |
| Project Details | POPR-QT-E-004 | High | The system shall provide a Project field to display the project associated with the selected purchase requisition. | Quotation Tracking Screen | Functional Testing, Data Validation | Purchase Requisition Tracking |
| Select PR to Track | POPR-QT-E-005 | High | The system shall provide a mandatory Select PR to Track dropdown to allow users to select a purchase requisition for quotation tracking. | Quotation Tracking Screen | Functional Testing, Validation Testing | Quotation Management |
| Purchase Requisition Options | POPR-QT-E-006 | High | The system shall display available purchase requisitions in the Select PR to Track dropdown, identifying each option by its Requirement No and associated Project. | Select PR to Track Dropdown | Functional Testing, Integration Testing | Purchase Requisition Tracking |
| Purchase Requisition Selection | POPR-QT-E-007 | High | The system shall allow users to select a purchase requisition from the available dropdown options for quotation tracking. | Quotation Tracking Screen | Functional Testing, UAT | Quotation Management |
| PR and Project Information | POPR-QT-E-008 | Medium | The system shall display the corresponding PR No. and Project information for the selected purchase requisition. | Quotation Tracking Screen | Functional Testing, Data Validation | Purchase Requisition Tracking |

### 3.3.B: Approved PRs – Accounts (POPR-APA-F)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Approved PRs – Accounts View List | POPR-APA-F-001 | High | The system shall display approved purchase requisitions in a centralized listing for the Accounts team to review and process for vendor assignment. | Approved PRs – Accounts Screen | Functional Testing, UAT | Purchase Requisition Management |
| Screen Description | POPR-APA-F-002 | Low | The system shall display a description indicating that approved purchase requisitions are ready for vendor assignment. | Approved PRs – Accounts Screen | Functional Testing | Navigation |
| Approved Purchase Requisition Listing | POPR-APA-F-003 | High | The system shall display approved purchase requisition records with Sl No, Requirement No, Project, Priority, Required Date, Requested By, Status, and Actions. | Approved Purchase Requisitions | Functional Testing, UAT | Purchase Requisition Management |
| Search Approved PRs | POPR-APA-F-004 | Medium | The system shall provide a search field to allow users to search approved purchase requisitions using supported search criteria. | Approved Purchase Requisitions | Functional Testing, Validation Testing | Search and Filtering |
| Record Count | POPR-APA-F-005 | Low | The system shall display the total number of approved purchase requisition records in the listing. | Approved Purchase Requisitions | Functional Testing | Purchase Requisition Management |
| Requisition Status | POPR-APA-F-006 | High | The system shall display the approval status of each purchase requisition in the listing. | Approved Purchase Requisitions | Functional Testing, Data Validation | Purchase Requisition Management |
| View Purchase Requisition | POPR-APA-F-007 | High | The system shall provide a View action for each approved purchase requisition to allow the Accounts team to review its details. | Approved Purchase Requisitions | Functional Testing, UAT | Purchase Requisition Management |
| Requisition Identification | POPR-APA-F-008 | High | The system shall display the Requirement No, Request Date, and Requested By details for the selected purchase requisition. | Purchase Requisition Details | Functional Testing, Data Validation | Request Information |
| Requester Information | POPR-APA-F-009 | Medium | The system shall display the requester's Designation, Department, and Reporting To details for the selected purchase requisition. | Purchase Requisition Details | Functional Testing | Request Information |
| Requisition Details | POPR-APA-F-010 | High | The system shall display the selected purchase requisition's Project, Priority, Required Date, and Purpose / Justification. | Purchase Requisition Details | Functional Testing, UAT | Request Information |

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| Requested Items | POPR-APA-F-011 | High | The system shall display the items associated with the selected purchase requisition, including the serial number, Item Name, and Qty. | Purchase Requisition Details – Items | Functional Testing, Data Validation | Item Management |
| Quotation Details | POPR-APA-F-012 | High | The system shall display the quotations associated with the selected purchase requisition, including the vendor name, quotation reference number, attached quotation file, and available quotation amount details. | Purchase Requisition Details – Quotations | Functional Testing, Integration Testing | Quotation Management |
| Supporting Attachments | POPR-APA-F-013 | Medium | The system shall display supporting attachments associated with the selected purchase requisition and provide access to the available files. | Purchase Requisition Details – Attachments | Functional Testing, File Validation | Document Management |
| Approval History | POPR-APA-F-014 | High | The system shall display the approval history of the selected purchase requisition, including the user involved, action or status description, and corresponding date and time. | Purchase Requisition Details – Approval History | Functional Testing, UAT | Approval Management |
| Vendor Assignment | POPR-APA-F-015 | High | The system shall provide an Assign action for each approved purchase requisition to initiate the vendor assignment process. | Approved Purchase Requisitions | Functional Testing, UAT | Vendor Management |
| Vendor Assignment | POPR-APA-F-015 | High | Upon clicking the Assign button, the system should redirect to the “Vendor Assignment & Split” Screen - POPR-VAS-G | POPR-VAS-G | Functional Testing, UAT | Navigation |
| Close Requisition Details | POPR-APA-F-017 | Medium | The system shall provide a close option in the purchase requisition details view to allow users to close the details and return to the approved purchase requisition listing. | Purchase Requisition Details | Functional Testing | Navigation |

### 3.3.B: Vendor Assignment & Split (POPR-VAS-G)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Vendor Assignment & Split – Screen | POPR-VAS-G-001 | High | The system shall provide a Vendor Assignment & Quantity Split screen to allow authorised users to assign approved purchase requisition quantities to one or more vendors. | Vendor Assignment & Split Screen | Functional Testing, UAT | Vendor Management |

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| Screen Description | POPR-VAS-G-002 | Low | The system shall display a description indicating that approved quantities can be assigned to one or more vendors. | Vendor Assignment & Split Screen | Functional Testing | Navigation |
| Workflow Steps | POPR-VAS-G-003 | High | The system shall display the workflow steps Assign Vendors & Qty, Enter Pricing, and Review & Generate to indicate the stages of the vendor assignment process. | Vendor Assignment & Split Screen | Functional Testing, UAT | Procurement Workflow |
| View PR Details | POPR-VAS-G-004 | Medium | The system shall provide a View PR Details option to allow users to access the details of the selected purchase requisition. | Vendor Assignment & Split Screen | Functional Testing | Purchase Requisition Management |
| PR Number | POPR-VAS-G-005 | High | The system shall display the PR No. associated with the selected purchase requisition. | Vendor Assignment & Split Screen | Functional Testing, Data Validation | Purchase Requisition Management |
| Project Details | POPR-VAS-G-006 | High | The system shall display the Project associated with the selected purchase requisition. | Vendor Assignment & Split Screen | Functional Testing, Data Validation | Purchase Requisition Management |
| Select PR to Assign | POPR-VAS-G-007 | High | The system shall provide a Select PR to Assign dropdown to allow users to select an approved purchase requisition for vendor assignment. | Vendor Assignment & Split Screen | Functional Testing, Validation Testing | Purchase Requisition Management |
| Approved Item Quantities | POPR-VAS-G-008 | High | The system shall display the approved quantities for each item in a read-only table containing Sl No, Item Description, UOM, and Approved Qty. | Item-wise Approved Quantity | Functional Testing, Data Validation | Item Management |
| Item Selection | POPR-VAS-G-009 | High | The system shall provide a Select Item dropdown to allow users to choose an item from the selected purchase requisition for vendor assignment. | Assign Vendors & Split Quantity | Functional Testing, Validation Testing | Item Management |
| Vendor Selection | POPR-VAS-G-010 | High | The system shall provide a Vendor Name dropdown for each assignment row to allow users to select a vendor for the selected item. | Assign Vendors & Split Quantity | Functional Testing, Integration Testing | Vendor Management |
| Available Approved Quantity | POPR-VAS-G-011 | High | The system shall display the available approved quantity for the selected item to indicate the quantity available for vendor assignment. | Assign Vendors & Split Quantity | Functional Testing, Data Validation | Quantity Management |
| Assign Quantity | POPR-VAS-G-012 | High | The system shall provide an Assign Qty field to allow users to specify the quantity allocated to each selected vendor. | Assign Vendors & Split Quantity | Functional Testing, Validation Testing | Quantity Management |
| Balance Quantity | POPR-VAS-G-013 | High | The system shall calculate and display the balance quantity remaining after vendor quantities are assigned. | Assign Vendors & Split Quantity | Functional Testing, | Quantity Management |

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| | | | | | Calculation Testing | |
| Add Vendor | POPR-VAS-G-014 | High | The system shall provide an Add Vendor option to allow users to add additional vendor assignment rows for the selected item. | Assign Vendors & Split Quantity | Functional Testing, UAT | Vendor Management |
| Edit Vendor Assignment | POPR-VAS-G-015 | Medium | The system shall provide an Edit action to allow users to modify an existing vendor assignment or its assigned quantity. | Assign Vendors & Split Quantity | Functional Testing, UAT | Vendor Management |
| Remove Vendor Assignment | POPR-VAS-G-016 | Medium | The system shall provide an action to remove a vendor assignment row before the assignment is finalized. | Assign Vendors & Split Quantity | Functional Testing | Vendor Management |
| Total Assigned Quantity | POPR-VAS-G-017 | High | The system shall calculate and display the total quantity assigned to vendors for the selected item. | Assign Vendors & Split Quantity | Functional Testing, Calculation Testing | Quantity Management |
| Remaining Quantity | POPR-VAS-G-018 | High | The system shall display the remaining quantity by comparing the approved quantity with the total assigned quantity. | Assign Vendors & Split Quantity | Functional Testing, Calculation Testing | Quantity Management |
| Quantity Validation | POPR-VAS-G-019 | High | The system shall validate vendor-assigned quantities against the approved quantity and indicate any quantity remaining to be assigned. | Assign Vendors & Split Quantity | Validation Testing, Functional Testing | Quantity Management |
| Clear Assignment | POPR-VAS-G-020 | Medium | The system shall provide a Clear option to clear the current vendor assignment entries for the selected item. | Assign Vendors & Split Quantity | Functional Testing | Form Management |
| Save & Next Item | POPR-VAS-G-021 | High | The system shall provide a Save & Next Item option to save the current item's vendor assignment details and proceed to the next item. | Assign Vendors & Split Quantity | Functional Testing, Data Persistence Testing | Vendor Management |
| Overall Assignment Summary | POPR-VAS-G-022 | High | The system shall display an Overall Assignment Summary containing Sl No, Item, Total Approved Qty, Total Assigned Qty, Status, and Action for each item. | Overall Assignment Summary | Functional Testing, UAT | Assignment Management |
| Assignment Status | POPR-VAS-G-023 | High | The system shall display the assignment status for each item, including the quantity remaining to be assigned. | Overall Assignment Summary | Functional Testing, Data Validation | Assignment Management |
| Edit Item Assignment | POPR-VAS-G-024 | Medium | The system shall provide an Edit action in the Overall Assignment Summary to | Overall Assignment Summary | Functional Testing, UAT | Assignment Management |

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| | | | allow users to modify the vendor assignment details for a selected item. | | | |
| Attach Quotation | POPR-VAS-G-025 | High | The system shall provide an Attach Quotation dropdown to allow users to select an available quotation for association with the vendor assignment. | Vendor Assignment & Split Quantity | Functional Testing, Integration Testing | Quotation Management |
| Selected Item Summary | POPR-VAS-G-026 | Medium | The system shall display the selected Item, UOM, and Approved Qty in the vendor assignment section to identify the item being allocated. | Vendor Assignment & Split Quantity | Functional Testing, Data Validation | Item Management |
| Proceed to Pricing | POPR-VAS-G-027 | High | The system shall provide a Proceed to Pricing option to allow users to continue to the pricing stage of the vendor assignment workflow. | Overall Assignment Summary | Functional Testing, UAT | Procurement Workflow |

### 3.3.B: Generated PO List (POPR-GPOL-H)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Generated PO List – Screen | POPR-GPOL-H-001 | High | The system shall provide a Generated Purchase Orders screen to allow authorized users to view, edit, delete, and print generated Purchase Orders and Office Notes. | Generated PO List Screen | Functional Testing, UAT | Purchase Order Management |
| Screen Description | POPR-GPOL-H-002 | Low | The system shall display a description indicating that users can view, edit, delete, and print purchase requisition Office Notes and Purchase Order documents. | Generated PO List Screen | Functional Testing | Navigation |
| Purchase Order Listing | POPR-GPOL-H-003 | High | The system shall display generated purchase orders in a centralized table containing Sl No, PO Number, PR Number, Vendor, Project, Total Items, Total Qty, Grand Total, Status, and Actions. | PO List | Functional Testing, UAT | Purchase Order Management |
| Search Purchase Orders | POPR-GPOL-H-004 | Medium | The system shall provide a search field to allow users to search generated purchase orders using supported search criteria. | PO List | Functional Testing, Validation Testing | Search and Filtering |
| Record Count | POPR-GPOL-H-005 | Low | The system shall display the total number of purchase order records in the listing. | PO List | Functional Testing | Purchase Order Management |
| Purchase Order Status | POPR-GPOL-H-006 | High | The system shall display the status of each generated purchase order in the listing. | PO List | Functional Testing, Data Validation | Purchase Order Management |

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| View Purchase Order | POPR-GPOL-H-007 | High | The system shall provide a View action for each generated purchase order to allow users to review its details. | PO List – View Action | Functional Testing, UAT | Purchase Order Management |
| PO Identification Details | POPR-GPOL-H-008 | High | The system shall display the PO Number, PR Number, Project, Vendor, Vendor Code, and Created Date for the selected purchase order. | Purchase Order Details | Functional Testing, Data Validation | Purchase Order Management |
| Purchase Order Item Details | POPR-GPOL-H-009 | High | The system shall display the items associated with the selected purchase order, including Sl No, Item Name, UOM, Qty, Unit Rate, GST %, Amount, Tax, and Total. | Purchase Order Details – Items | Functional Testing, Calculation Testing | Item and Pricing Management |
| Purchase Order Totals | POPR-GPOL-H-010 | High | The system shall display the total amount, total tax amount, and Grand Total for the selected purchase order based on the listed item details. | Purchase Order Details – Grand Total | Functional Testing, Calculation Testing | Financial Management |
| Office Note Generation | POPR-GPOL-H-011 | High | The system shall provide an Office Note action to display the Office Note document associated with the purchase requisition. | PO List – Office Note Action | Functional Testing, Document Validation | Document Management |
| Office Note Details | POPR-GPOL-H-012 | High | The system shall generate or display an Office Note containing the company name, submitted-by details, submitted-to details, reference number, date, subject, proposal, background, and item-wise financial details. | Office Note Document | Functional Testing, Document Validation, UAT | Document Management |
| Print Office Note | POPR-GPOL-H-013 | Medium | The system shall provide a Print option to allow users to print the displayed Office Note document. | Office Note Document | Functional Testing, Print Testing | Document Management |
| Purchase Order Document | POPR-GPOL-H-014 | High | The system shall provide a PO Doc action to display the generated Purchase Order document associated with the selected purchase order. | PO List – PO Doc Action | Functional Testing, Document Validation | Document Management |
| Purchase Order Document Details | POPR-GPOL-H-015 | High | The system shall display the Purchase Order document with the PO reference, date, vendor details, delivery or supply description, item quantities, unit rates, GST, tax amounts, totals, and applicable terms and conditions. | Purchase Order Document | Functional Testing, Document Validation, UAT | Document Management |
| Print Purchase Order | POPR-GPOL-H-016 | Medium | The system shall provide a Print option to allow users to print the generated Purchase Order document. | Purchase Order Document | Functional Testing, Print Testing | Document Management |

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| Edit Purchase Order | POPR-GPOL-H-017 | High | The system shall provide an Edit action to allow authorized users to modify the editable details of an existing purchase order. | PO List – Edit Action | Functional Testing, UAT | Purchase Order Management |
| Editable Item Details | POPR-GPOL-H-018 | High | The system shall allow users to edit the quantity, unit rate, and GST percentage for each item in the purchase order editing form. | Edit Purchase Order – Editable Items | Functional Testing, Validation Testing | Item and Pricing Management |
| Automatic Amount Calculation | POPR-GPOL-H-019 | High | The system shall calculate and display the item amount, tax amount, and total amount based on the entered quantity, unit rate, and GST percentage. | Edit Purchase Order – Editable Items | Functional Testing, Calculation Testing | Financial Management |
| Updated PO Totals | POPR-GPOL-H-020 | High | The system shall update the total quantity, total amount, total tax, and Grand Total when the editable item details are changed. | Edit Purchase Order | Functional Testing, Calculation Testing | Financial Management |
| Save Purchase Order Changes | POPR-GPOL-H-021 | High | The system shall provide a Save Changes option to save modifications made to the purchase order. | Edit Purchase Order | Functional Testing, Data Persistence Testing | Purchase Order Management |
| Cancel Editing | POPR-GPOL-H-022 | Medium | The system shall provide a Cancel option to exit the purchase order editing form without saving the pending changes. | Edit Purchase Order | Functional Testing | Navigation |
| Delete Purchase Order | POPR-GPOL-H-023 | High | The system shall provide a Delete action to allow authorized users to initiate deletion of a generated purchase order. | PO List – Delete Action | Functional Testing, UAT | Purchase Order Management |
| Confirm Delete PO | POPR-GPOL-H-024 | High | The system shall display a confirmation dialog identifying the PO Number and Vendor before deleting the selected purchase order. | Confirm Delete PO Dialog | Functional Testing, Validation Testing | Purchase Order Management |
| Confirm or Cancel Deletion | POPR-GPOL-H-025 | High | The system shall provide Delete PO and Cancel options in the confirmation dialog to allow users to confirm the deletion or cancel the operation. | Confirm Delete PO Dialog | Functional Testing, UAT | Purchase Order Management |
| Close Document View | POPR-GPOL-H-026 | Medium | The system shall provide a close option in the Purchase Order details, Office Note, and PO document views to allow users to return to the Generated PO List. | Document View Screens | Functional Testing | Navigation |

### 3.3.B: Document Upload & Tracking (POPR-DUT-I)

| Screen / Feature | Req ID | Priority | Requirement Description | Reference | Verification & Validation | Hierarchy |
|---|---|---|---|---|---|---|
| Document Upload & Tracking – Screen | POPR-DUT-I-001 | High | The system shall provide a Document Upload & Tracking screen to allow users to upload and track procurement documents associated with purchase requisitions for the Accounts team. | Document Upload & Tracking Screen | Functional Testing, UAT | Document Management |
| Screen Description | POPR-DUT-I-002 | Low | The system shall display a description indicating that users can upload and track procurement documents for the Accounts team. | Document Upload & Tracking Screen | Functional Testing | Navigation |
| Total Documents | POPR-DUT-I-003 | Medium | The system shall display the total number of procurement documents across all purchase requisitions. | Document Status Summary | Functional Testing, Data Validation | Document Management |
| Verified Documents | POPR-DUT-I-004 | High | The system shall display the count of verified documents that are ready for payment processing. | Document Status Summary – Verified | Functional Testing, Data Validation | Document Verification |
| Pending Review Documents | POPR-DUT-I-005 | High | The system shall display the count of documents awaiting verification or review. | Document Status Summary – Pending Review | Functional Testing, Data Validation | Document Verification |
| Rejected Documents | POPR-DUT-I-006 | High | The system shall display the count of rejected documents that require re-upload. | Document Status Summary – Rejected | Functional Testing, Data Validation | Document Verification |
| PR Number | POPR-DUT-I-007 | High | The system shall provide a PR No. field to display the purchase requisition number associated with the selected requisition. | Document Upload & Tracking Screen | Functional Testing, Data Validation | Purchase Requisition Tracking |
| Project Details | POPR-DUT-I-008 | High | The system shall provide a Project field to display the project associated with the selected purchase requisition. | Document Upload & Tracking Screen | Functional Testing, Data Validation | Purchase Requisition Tracking |
| Select PR to Track | POPR-DUT-I-009 | High | The system shall provide a mandatory Select PR to Track dropdown to allow users to select a purchase requisition for document tracking. | Document Upload & Tracking Screen | Functional Testing, Validation Testing | Document Management |
| Purchase Requisition Options | POPR-DUT-I-010 | High | The system shall display available purchase requisitions in the selection dropdown, identifying each option by its Requirement No and associated Project. | Select PR to Track Dropdown | Functional Testing, Integration Testing | Purchase Requisition Tracking |

| | | | | | | |
|---|---|---|---|---|---|---|
| Purchase Requisition Selection | POPR-DUT-I-011 | High | The system shall allow users to select a purchase requisition from the available dropdown options for document tracking. | Document Upload & Tracking Screen | Functional Testing, UAT | Document Management |
| PR and Project Information | POPR-DUT-I-012 | Medium | The system shall display the corresponding PR No. and Project information for the selected purchase requisition. | Document Upload & Tracking Screen | Functional Testing, Data Validation | Purchase Requisition Tracking |

## 3.3.B1: Field Executive Module - Screen: Add Candidate – Field Validation Rules (SRS)

| Screen | Field Name | Validation Requirement | Mandatory / Optional | Remarks |
|---|---|---|---|---|
| Login | Employee ID | The system shall accept a valid Employee ID as configured in the User Master. Only active users shall be allowed to log in. (8-Digit ID) | Mandatory | Numeric format as defined by the organization. |
| Login | Password | The system shall accept the password associated with the Employee ID. Password shall be masked during entry. | Mandatory | Password Validation |
| Login | Login Button | The system shall authenticate the entered Employee ID and Password. Upon successful authentication, the user shall be redirected to the dashboard based on assigned role. | Mandatory | Session shall be created after successful authentication. |
| Login | Error Message | The system shall display an appropriate error message for invalid Employee ID or Password without revealing which field is incorrect. | System Generated | To prevent information disclosure. |
| Add Candidate | Aadhaar Upload | The system shall allow upload of either **one PDF** or up to **three images** for Aadhaar document, with at **least one image mandatory**. | Mandatory | Additional upload rules defined in Design Document |
| Add Candidate | Aadhaar Number | The system shall accept a valid 12-digit numeric Aadhaar number. | Mandatory | OCR auto-fill allowed with manual edit |
| Add Candidate | Mobile Number | The system shall accept a valid 10-digit numeric mobile number. | Mandatory | Used for candidate communication |
| Add Candidate | Alternate Phone Number | The system shall accept a valid numeric mobile number. | Optional | — |
| Add Candidate | Skill Category | The system shall allow selection of a skill category (Civil or Mechanical) | Mandatory | Civil / Mechanical |
| Add Candidate | Trade | System shall dynamically display relevant Trade options based on the selected skill category | Mandatory | Skill master maintained in system |
| Add Candidate | Experience | The system shall accept numeric input including decimal values for experience. | Mandatory | Value captured in years |
| Add Candidate | Candidate Photo | The system shall allow capture of candidate photo using device camera. | Mandatory | Live Photo |
| Add Candidate | UAN Number | The system shall allow capture of UAN number as an optional field. | Optional | — |
| Add Candidate | Preferred Location | The system shall allow selection of preferred work location. | Mandatory | — |
| Add Candidate | Tentative Joining Date | The system shall allow selection of a tentative joining date. | Mandatory | Calendar-based input |

### 3.3.B2: Module Dependency & Data Flow (Field Executive → Site Executive Dependency)

| Ref ID | Dependency Description | Impacted Module | Trigger Condition | Expected System Behavior |
|---|---|---|---|---|
| DEP-01 | Candidates uploaded by Field Executive shall be available for onboarding by Site Executive. | FE → SE | After successful upload by FE | Candidate record shall appear in SE "New Entry / Onboarding" module. |
| DEP-02 | Total Onboarded count in Field Executive dashboard depends on successful onboarding by Site Executive. | SE → FE | Gate Pass marked as "Yes" during verification | System shall increment "Total Onboarded" metric in respective Field Executive dashboard. |
| DEP-03 | If candidate moves to Verification Exit in SE module, it shall not be counted under Total Onboarded in FE dashboard. | SE → FE | Any verification marked as "No" | Candidate status updated as Verification Exit and excluded from FE Onboarded KPI. |
| DEP-04 | Verification-in-Progress candidates shall not be counted as Onboarded in FE dashboard. | SE → FE | Verification submitted but Gate Pass not issued | Candidate reflected under Verification-in-Progress only. |

## 3.4 Operating Environment Requirements

This section outlines the necessary hardware, software, network, and communication tools required for the successful deployment, operation, and maintenance of the Document Management Software across Power Transmission offices (BESCOM, HESCOM, GESCOM) and associated units

### 3.4.1 Hardware

**Server-Side (for Hosting Application & Database):**

- **Processor:** Minimum 8-core CPU (Intel Xeon / AMD EPYC) for handling multiple simultaneous uploads, versioning processes, and search indexing.
- **RAM:** 32 GB or higher to support concurrent user sessions, document preview generation, and background processing (e.g., virus scans, OCR).
- **Storage:** 2 TB SSD (expandable) with RAID support to ensure fast retrieval and redundancy for high-volume document storage.
- **Backup:** External backup device or cloud backup service configured for daily incremental and weekly full backups.
- **UPS:** Power backup to ensure uninterrupted DMS services.
- **Environment:** Rack-mounted in a secured on-premises data center or hosted on a compliant cloud platform.

**Client-Side (Office Users):**

- **Processor**: Intel i5 or higher.
- **RAM**: 8 GB minimum to ensure smooth document preview and editing operations.
- **Storage**: 256 GB SSD or higher.
- **Display**: 15-inch Full HD or higher resolution for clear viewing of scanned documents.
- **Peripheral Devices:**
- **Scanner** – High-resolution (minimum 300 dpi) for digitizing paper-based documents.
- **Printer** – For printing approval/rejection reports, workflow summaries, or document metadata sheets.

### 3.4.2 Software

**Server-Side:**

- **Operating System:** Windows Server 2019 / Ubuntu Linux 20.04 or later
- **Database**: MS SQL Server / MySQL
- **Application Stack:**
    - **Backend:** NodeJS/ExpressJS.MySql
    - **Frontend:** ReactJS /Bootstrap5
- **Web Server:** AWS/ IIS
- **Security Tools:** SSL Certificate, Endpoint Protection

**Client-Side (Office Users):**

- **Web Browser:** Chrome / Edge (latest version)
- PDF Viewer, Office Suite (MS Office or LibreOffice)

### 3.4.3 Network

**Internal Network:** LAN/WAN within offices

**Internet Connectivity:**

- Office: Minimum 10 Mbps broadband per location
- Field: 4G data connectivity for mobile app users
- VPN (if remote access is needed from outside premises)
- Firewall and Network Security Appliances in place

### 3.4.4 Communication

- Inter-Office Communication:

The system shall support seamless and secure data exchange between headquarters, regional offices, and field units to ensure smooth operational coordination.

- User Notifications:

The system shall provide automated notifications through email, SMS, and in-application alerts for key events including document uploads, approvals, rejections, status changes, and version updates.

- Collaboration & Review Communication:

The system shall provide built-in collaboration features such as comments, annotations, and review notes to facilitate effective communication between users involved in document submission, verification, and approval processes.

- Helpdesk & Support Communication:

The system shall provide designated support channels such as phone, email, or integrated support modules to allow users to raise issues or request operational assistance.

- Audit & Reporting Communication:

The system shall support scheduled and on-demand distribution of reports to authorized users for audit purposes, workflow tracking, compliance monitoring, and document version history.

## 3.5 External interface requirements

This section describes the external interfaces of the system, including user interfaces, hardware interfaces, software interfaces, and data interfaces. It outlines how the system interacts with users, external systems, and supporting infrastructure, along with key operational and performance considerations for each interface.

### 3.5.1 User Interface (UI) Requirements – WFMS

The system shall provide web-based and mobile-based user interfaces appropriate to each user role. Interfaces shall be intuitive, role-driven, and accessible to authorized users for performing assigned tasks.

### 3.5.2 Hardware Interface Requirements – WFMS

The system shall support standard computing devices including desktops, laptops, and mobile devices. Mobile application functionality shall leverage device capabilities such as camera and GPS where required.

### 3.5.3 Software Interface

The system shall interface with external software systems such as authentication services, messaging gateways (SMS/email), and third-party verification services as required for workflow execution.

#### 3.5.4 Communication Interfaces

The system shall support secure communication over standard internet protocols to enable data exchange between clients, servers, and integrated external systems.

#### 3.5.5 Operational Requirements

This section defines the operational constraints and performance expectations under which the system shall function to support business operations effectively.

##### 3.5.5.1 Performance Requirements

- The system shall support concurrent access by multiple users without degradation of core functionalities.
- User authentication and dashboard loading operations shall complete within acceptable response times under normal operating conditions.
- Data submission actions such as candidate onboarding, attendance marking, and verification updates shall be processed reliably and within defined response thresholds.
- The system shall be scalable to accommodate increased data volume and user load as operational usage grows.
- The system shall ensure consistent performance during peak operational hours.

### 3.6 Availability & Reliability Requirements

- The system shall be available for operational use during defined business hours, excluding scheduled maintenance windows.
- The system shall be designed to ensure reliable operation under normal and peak workload conditions.
- The system shall support graceful recovery in the event of application failure without loss of critical transactional data.
- Scheduled maintenance activities shall be planned to minimize disruption to business operations.
- The system shall maintain data consistency and integrity during unexpected interruptions

### 3.7 Security & Compliance Requirements

- The system shall enforce role-based access control to ensure users can access only authorized functions and data.
- The system shall protect sensitive personal and identification data through secure storage and controlled access mechanisms.
- The system shall support secure authentication mechanisms for all users accessing web and mobile applications.
- The system shall maintain audit logs for critical actions including data creation, modification, verification, and approval.
- The system shall comply with applicable data protection and privacy regulations as required by organizational and statutory guidelines.
- The system shall ensure secure transmission of data between client applications and backend services

## 3.6 Non-Functional Requirements –Work Force Management System (DMS)

| Non-Functional Requirement | NFR-ID | Fit Criteria (WFMS) |
|---|---|---|
| **Performance** | NFR-PRE-001 | The system shall respond to standard user requests within **3 seconds** under normal operating conditions. |
| **Performance** | NFR-PER-002 | User authentication shall complete within **2 seconds** for 95% of requests. |
| **Performance** | NFR-PER-003 | Dashboard pages shall load within **3 seconds**. |
| **Scalability** | NFR-SCL-001 | The application shall support future horizontal scaling without functional changes. |
| **Scalability** | NFR-SCL-002 | The system shall support increasing workforce and candidate records without impacting core functionality. |
| **Security** | NFR-SEC-001 | All communication shall use HTTPS/TLS encryption. |
| **Security** | NFR-SEC-002 | Sensitive information shall be encrypted at rest. |
| **Security** | NFR-SEC-003 | Role-Based Access Control (RBAC) shall restrict access based on user roles |
| **Availability** | NFR-AVL-001 | The system shall maintain at least **95% availability**, excluding planned maintenance. |
| **Availability** | NFR-AVL-002 | Scheduled maintenance shall be communicated in advance. |
| **Compatibility** | NFR-COM-001 | The web application shall function correctly on the latest versions of major browsers (Chrome, Firefox), and the mobile application shall support standard Android devices used in field operations. |
| **Compatibility** | NFR-COM-002 | Mobile interfaces shall support Android version X and above (if applicable). |
| **Backup & Recovery** | NFR-BKP-001 | The database shall be backed up weekly |
| **Backup & Recovery** | NFR-BKP-002 | Backup data shall be retained according to organizational policy. |

### 3.6.1 Security & Access Control

**a. Role-Based Access Control (RBAC)**

The system shall support role-based access control with multiple user roles such as Admin, Coordinator, Site Executive, Field Executive, and Auditor. Each role shall be assigned specific permissions to control access to system functions and data.

Fit Criteria:
Verify users can access only role-authorized functions during UAT.

**b. Multi-Factor Authentication**

**The system shall enforce multi-factor authentication for privileged roles such as Admin and Coordinator using an additional verification mechanism beyond username and password.**

**Fit Criteria:**
**Verify additional authentication is required for configured roles.**

**c. Data Access Restrictions**

The system shall restrict users to view, edit, or download only those records and documents for which they are authorized based on role and workflow stage.

**Fit Criteria:**
Verify unauthorized access to restricted data is blocked.

**d. Sensitive Data Protection**

Access to sensitive personal and identification data shall require explicit authorization as per system configuration.

**Fit Criteria:**
Verify restricted data access requires appropriate permissions.

**e. Encryption & Secure Communication**

The system shall ensure secure storage and transmission of sensitive data using industry-standard encryption mechanisms.

**Fit Criteria:**
Verify data is encrypted at rest and during transmission.

**f. Audit Trails**

The system shall maintain audit logs for critical user actions including data creation, modification, verification, approval, and deletion. Audit logs shall include user identification, timestamp, and action performed.

**Fit Criteria:**
Verify audit logs are generated and retrievable for compliance review.

**g. Security Event Alerts**

The system shall generate alerts for suspicious activities such as repeated failed login attempts or unauthorized access attempts.

**Fit Criteria:**
Verify security alerts are triggered for configured events.

**h. Server & Infrastructure Security**

The system shall be deployed in a secure server environment with controlled access, malware protection, and network security measures.

**Fit Criteria:**
Verify infrastructure security controls are in place.

**i. Single Sign-On (Future Ready)**

The system shall be designed to support integration with enterprise single sign-on (SSO) solutions in future phases.

**Fit Criteria:**
Verify SSO integration capability through system design documentation.

### 3.6.2 Maintainability

- The system shall be designed using a modular architecture to support ease of maintenance and future enhancements.
- The system shall provide adequate technical documentation including architecture overview, database design, API references, configuration, and operational procedures.
- The system shall follow secure coding practices and industry-recommended development standards.
- Version control and change history shall be maintained for all major system updates.

### 3.6.3 Availability

**Availability**

- The system shall be available for operational use with minimal downtime, excluding scheduled maintenance.
- Planned maintenance activities shall be communicated to stakeholders in advance.
- The system shall support recovery from failures within a reasonable time frame to ensure business continuity.

**Scalability**

- The system shall be scalable to support growth in the number of users, records, and transactions without significant performance degradation.
- The system architecture shall support horizontal and vertical scaling as required by operational demand.

### 3.6.4 Portability

- The system shall be accessible through commonly used web browsers on desktop and laptop devices.
- The system shall support deployment on standard server operating systems.
- The system shall support access from mobile devices through responsive web design and/or mobile applications.

### 3.6.5 Legal / Regulatory

- The system shall comply with applicable information technology laws, data protection regulations, and organizational policies.

- The system shall support retention and archival of records as per regulatory and business requirements.
- The system shall ensure integrity and authenticity of stored records for audit and compliance purposes Data Migration

**3.6.6 Data Migration**

- The system shall support migration of existing data from legacy sources into the new system.
- Data migration shall include validation to ensure accuracy and completeness of migrated records.
- Rollback mechanisms shall be available to restore data in case of migration issues.

**3.6.7 Installation & Deployment**

- The system shall provide installation and deployment guidelines for supported environments.
- The system shall support deployment in on-premise and cloud environments.
- Separate environments for development, testing, and production shall be maintained.

**3.6.8 User Training**

- The vendor shall provide user training for different user roles prior to system rollout.
- Training materials such as manuals or reference guides shall be made available to users.
- Post-deployment support shall be provided during the initial rollout phase.

**3.6.9 User Manual & Help**

- The system shall provide a user manual covering core functionalities and workflows.
- Context-sensitive help and guidance shall be available within the application.
- Frequently asked questions (FAQs) and basic troubleshooting guidance shall be provided.

### 3.6.6 User Manual and Help

- The system shall provide a comprehensive user manual covering key system functions including navigation, document upload and search, metadata management, and approval workflows.
- The system shall provide context-sensitive help within the user interface to guide users during system operations.
- The system shall include a Frequently Asked Questions (FAQ) section and “How-to” guides to assist users in performing common tasks.
- User help documentation shall be accessible to authorized users within the application.

### 3.6.7 Non-Functional Testing for DMS

1. Performance Testing
    - Performance testing shall be conducted to validate system response times under normal and peak load conditions.

- Load and stress testing shall be performed to assess system behavior with increasing numbers of concurrent users.
- Bulk data operations shall be tested to ensure acceptable performance during large data uploads and retrievals.

2. Security Testing
    - Security testing shall be conducted to verify role-based access control, authentication mechanisms, and data access restrictions.
    - Vulnerability assessments shall be performed to identify common security weaknesses.
    - Audit logging and security alert mechanisms shall be tested to ensure proper tracking and notification of security events.
3. Maintainability Testing
    - Code reviews shall be performed to ensure adherence to secure coding and development standards.
    - API documentation and system documentation shall be reviewed to ensure accuracy and completeness after system changes.
    - Deployment and upgrade procedures shall be tested to validate ease of maintenance.
4. Scalability Testing
    - Scalability testing shall be conducted to evaluate system behavior under increasing user load.
    - The system shall be tested to ensure it can handle growth in data volume without significant degradation in performance.
    - The system architecture shall be evaluated for its ability to scale horizontally and/or vertically to meet future operational demands.
5. Portability Testing
    - The system shall be tested across supported browsers and operating systems to ensure consistent behavior.
    - Mobile accessibility and responsive behavior shall be validated for supported devices.
6. Data Migration Testing
    - Data migration testing shall be conducted to verify accuracy, completeness, and consistency of migrated data.
    - Rollback procedures shall be tested to ensure data can be restored in case of migration issues.
7. Automation vs Manual Testing
    - Automated testing shall be used for repetitive and system-level validations such as logging, backups, and recovery.
    - Manual testing shall be conducted for workflow-driven and user-interactive functionalities.

### 3.6.8 Features Not Required

The following features are explicitly excluded from the scope of the current phase of the system implementation to ensure clarity, focus, and controlled delivery. These items may be considered in future phases based on business requirements.

- The system shall not include online payment gateway integration or any payment processing functionality.
- The system shall support operation in English language only; multilingual or localization support is excluded in the current phase.
- The system shall not implement biometric authentication mechanisms such as fingerprint, facial recognition, or iris scanning.
- The system shall be accessible only to authorized internal users; no public or external user access portals shall be provided.
- The system shall not include AI or machine learning–based features such as automated document classification, tagging, or OCR processing.
- Integration with GIS or map-based systems is excluded from the current phase.
- The system shall not include a native or hybrid mobile application; access shall be provided through a web-based responsive interface only.
- Offline functionality for document upload, review, or processing shall not be supported in the current phase.
- The system shall not integrate with third-party cloud storage services such as Google Drive, Dropbox, or OneDrive.

# 4 Prototype

A functional prototype was developed to demonstrate and validate key system functionalities and workflows. The prototype was used primarily for requirement validation and stakeholder review.

**Prototype Features**

The prototype demonstrated the following core functionalities:

- Document upload with metadata capture
- Review and approval workflow
- Document version management
- Search, filter, and retrieval of documents
- Document activity history and audit trail view

**Modules Demonstrated**

The following modules were showcased in the prototype:

- Role-based dashboard views for Admin, Uploader, Reviewer, and Approver
- Document upload and validation screen
- Review and approval interface with comments
- Document version history view
- Advanced search with filters such as category, tags, and date

**Prototype Purpose**

The prototype was developed to:

- Validate business workflows and functional requirements
- Demonstrate user experience and navigation flow
- Collect feedback from stakeholders
- Identify gaps or missing requirements at an early stage

**Prototype Limitations**

The prototype was intended for demonstration purposes only and did not include:

- Complete backend integration with the production database
- Full security implementation or role-based authentication enforcement

# 5 Acceptance

## 5.1 Acceptance Criteria

The system shall be considered accepted by the customer upon successful completion of the following criteria:

**1. Functional Completion**

- All core modules shall be implemented in accordance with the approved functional requirements.
- Implemented functionalities shall align with the validated prototype and agreed workflows.

**2. Role-Based Access Control**

- Authorized user roles shall be able to log in and access only the functions and data permitted to their assigned roles.
- Role permissions shall be implemented as per the approved access control definitions.

**3. End-to-End Workflow Execution**

- The system shall support the complete document lifecycle, including:
  - Document upload with metadata capture
  - Review and approval or rejection with comments
  - Version management for updated documents
  - Final archival and retrieval of approved documents

**4. Data Accuracy and Integrity**

- Stored data and associated documents shall be accurate and consistent across all system modules.
- Workflow status, document versions, and audit information shall remain synchronized.

**5. UI/UX Consistency**

- The system shall provide a consistent and user-friendly interface across all modules.
- Screen layouts and navigation shall align with the approved prototype and usability guidelines.

**6. Performance and Load Handling**

- The system shall demonstrate acceptable performance for key operations under normal operating conditions.
- The system shall support concurrent user access without degradation of core functionality.

**7. Successful UAT (User Acceptance Testing)**

- User Acceptance Testing shall be conducted by the client or authorized representatives.
- All critical functional scenarios shall be validated, and identified high-severity issues shall be resolved prior to acceptance.

**8. Report Generation**

- The system shall support generation of operational and audit-related reports.
- Reports shall reflect current and accurate system data.

**9. Audit and Traceability**

- The system shall maintain audit logs for significant user actions.
- Audit information shall be accessible only to authorized users.

**10. Deployment and Documentation**

- The system shall be deployed to the agreed environments as per the deployment plan.
- Required documentation, including system, operational, and user documentation, shall be delivered.

**11. Security Compliance**

- The system shall implement appropriate security measures to protect data access and transmission.
- Authentication and authorization mechanisms shall follow industry-recommended practices.

**12. Backup and Recovery**

- The system shall support backup and recovery mechanisms to protect system data.
- Recovery procedures shall ensure data integrity in the event of failures.

## 6 References to Other Documents

<Provide references to Use Cases, Data Flow Diagrams etc >

## 7 Abbreviations / Bibliography

| Abbreviations | Meaning |
|---|---|
| MTBF | Mean Time Between Failure |
| RDBMS | Relational Database Management System |
| ATM | Automatic Transfer Of Money |